Electricity Supply Shocks and Manufacturing: Evidence from the Fukushima Nuclear Accident

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We show that adverse shocks to electricity supply—an input accounting for only 2% of revenue—can cause substantial and persistent declines in industrial production. Using confidential establishment-level data and instrumenting with initial supply conditions, we find that shocks following the 2011 Fukushima nuclear accident reduced Japanese manufacturing electricity use by 6.6% and output by 1.7%, implying annual output losses of 4.9 trillion Japanese yen (50 billion US dollars). Establishments adjusted little through energy substitution, product switching, outsourcing, within-firm reallocation, or relocation. Our production function highlights two mechanisms that amplify output losses: establishments cannot easily substitute other inputs for electricity, and reduced electricity use leaves machinery and equipment temporarily idle. We quantify how electricity-market reforms, such as regional market integration, can mitigate these losses.

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1 Introduction

Modern economies run on electricity, but its supply cannot be taken for granted, even in advanced economies. Russia's war in Ukraine and recurrent conflict in the Middle East illustrate what the International Energy Agency describes as "significant fragilities in today's global energy system" (IEA, 2024). Natural disasters pose additional risks to electricity supply. As economies replace fossil fuels with clean electricity, the stakes rise: the more they depend on electricity, the more they stand to lose when electricity supply is disrupted. Despite this growing dependence, little is known about how adverse shocks to electricity supply affect the industrial sector, a major consumer of electricity.[1]

This paper uses the sweeping changes in Japan's electricity policy following the 2011 Fukushima nuclear accident to study how sustained, nationwide electricity supply shocks affect manufacturing production. On March 11, 2011, a magnitude 9.0 earthquake and subsequent tsunami struck the Fukushima Daiichi nuclear power plant, triggering "the worst civil nuclear disaster since the one at Chernobyl in 1986" (UNSCEAR, 2021). The resulting reactor meltdowns and radioactive releases sparked widespread concerns about nuclear safety. The accident marked "a major turning point" in Japan's energy policy (ANRE, 2022), as the country responded by progressively taking nuclear reactors offline. Nuclear power's share of peak-day electricity supply collapsed from 17.4% in 2010 to 1.4% in 2013 (METI, 2016) and recovered to only 2.9% by 2018. This loss of nuclear supply triggered nationwide electricity shortages on an extraordinary scale—"perhaps one of the most severe electricity shortfalls in history" (IEA, 2011).

Our empirical strategy draws on confidential establishment-level data from Japan's Census of Manufactures for 2005–2018 and wide regional variation in electricity supply shocks. Although nuclear generation declined nationwide, the resulting supply shortfalls differed sharply across Japan's nine electricity service territories. The Kansai service territory, for example, had relatively little reserve capacity before the accident and immediately lost nuclear generation that had accounted for about 26% of its peak electricity supply. Despite efforts to expand generation from other sources, the region had to reduce electricity demand by as much as 16% at times to avoid blackouts.[2] The government sought to achieve these demand reductions largely through moral suasion. Replacing nuclear power with other energy sources, particularly expensive thermal generation, also drove up electricity prices. In con-

[1] A few studies examine how changes in electricity taxes or tariffs affect firm performance in advanced economies (Gerster and Lamp, 2024; von Graevenitz and Rottner, 2024). Others study blackouts or greyouts in developing economies (Fisher-Vanden et al., 2015; Allcott et al., 2016; Cole et al., 2018; Gonzales et al., 2026). A broader literature studies oil and other energy shocks (Kilian, 2008; Hamilton, 2013).

[2] Electricity supply must meet demand in real time, including at its peak. Nuclear power provided approximately 26% of Kansai's peak-day supply in 2010.

trast, regions with nuclear shares below 10% or large reserve margins, such as Chugoku and Hokkaido, faced relatively mild electricity supply shocks. As such, we exploit this regional variation using instruments based on pre-accident nuclear reliance and reserve margins.

The electricity supply shocks caused substantial and persistent contractions in manufacturing activity. After 2013, when Japan's nuclear shutdown looked set to last, the shocks reduced electricity use by 6.6%, labor and materials use by 1.3% each, and output by 1.7%. These contractions persisted through 2018, seven years after the accident. The 1.7% reduction in output implies an annual loss of approximately 4.9 trillion Japanese yen (50 billion US dollars), equivalent to roughly three years of manufacturing growth at the sector's pre-accident average annual real growth rate. Neither direct damage from the earthquake and nuclear accident nor supply-chain disruptions nor reductions in local demand can explain these effects on manufacturing.

The shocks induce little adjustment along margins that could mitigate output losses. Electricity supply shocks do not increase establishments' use of other energy inputs in production. The shocks also have little effect on outsourcing or product switching, two ways establishments could reduce the electricity intensity of production. Nor do they meaningfully shift production toward establishments in less affected regions or induce establishments to relocate. Instead, they mainly reduce operating hours and scale back production. Nonetheless, our interpretation is that establishments act to limit output losses when feasible. The roughly 10% of establishments that own generators do, in fact, increase on-site electricity generation in response to the shocks.[3] For most establishments, however, options for mitigating output losses appear limited or costly.

These empirical findings raise two related questions: why can shocks to electricity—an input with a small expenditure share—generate such large output losses, and how can policies mitigate these losses? In our data, electricity expenditure accounts for only 2% of establishment revenue, yet a 6.6% reduction in electricity use is associated with a 1.7% reduction in output, far exceeding a standard factor-share benchmark.[4] We have demonstrated one contributing factor—establishments have limited ability to cushion the shocks—but this alone does not fully explain the magnitude of these output losses.

To explain such large output losses, we estimate a production function and use it to predict input and output responses to electricity supply shocks. We incorporate two mech-

[3]Our data distinguish energy used directly in production from fuel used for on-site electricity generation. We find that the shocks increased fuel use for on-site generation among establishments with self-generators. About 10% of establishments owned generators before the shocks, and these establishments experienced no detectable output reductions.

[4]Under the factor-share benchmark (Hulten, 1978), the direct output effect is approximately the input's expenditure share times its percentage change. Holding other inputs fixed, a 6.6% decline in electricity use therefore implies an output loss of $0.02 \times 6.6\% \approx 0.13\%$, far below the observed 1.7%.

anisms that amplify output losses: limited substitutability between electricity and other inputs and electricity-induced capital idleness. Limited substitutability makes it difficult to replace electricity with other inputs. Capital idleness operates differently: the shocks reduce the utilization of existing capital and the services it provides, even after accounting for substitution among inputs (Finn, 2000). For example, a steel plant may be unable to fully utilize its electricity-dependent furnaces when electricity availability declines or when shocks reduce operating hours. The resulting capital idleness directly reduces output and indirectly amplifies the contraction by lowering the productivity of other inputs.

We estimate the model by matching its predicted responses of electricity use, labor, materials, and output to electricity supply shocks with the corresponding reduced-form estimates. Under a CES production technology, we estimate an elasticity of substitution of 0.57 between electricity and other inputs. If we instead assume greater substitutability, the model predicts substantially more substitution away from electricity. The model predicts that the shocks lead to a larger reduction in electricity use but smaller reductions in materials use and labor input, and therefore in output—a pattern inconsistent with the reduced-form estimates. We also estimate a capital-idleness elasticity of 0.17. In an average region, the shocks reduce electricity use by 6.6%. This leaves about 1.1% (6.6% $\times$ 0.17) of installed capital idle, reducing capital services by 1.1%. This decline in capital service may appear modest, yet it substantially amplifies output losses because capital plays a much larger role in production than electricity does directly. This capital idleness mechanism increases the predicted output loss from 0.6% to 1.7%, nearly tripling the losses.

How can policies mitigate the output losses caused by electricity supply shocks? We found that manufacturing establishments adapted little to the shocks, so we focus on policies that reduce the severity of the shocks they face. Japan used moral suasion to curb electricity demand, while replacing lost nuclear power with fossil-fuel generation raised electricity prices. Without moral suasion, our production function model still predicts that higher electricity prices reduce output by about 1.5%—roughly 90% of the combined impact. These results suggest that removing barriers to lower electricity prices in Japan could substantially mitigate output losses.

We quantify how electricity-market reforms, particularly regional market integration, can mitigate these losses by lowering electricity prices. Japan has long recognized that its fragmented electricity market creates inefficiencies (ANRE, 2017). Indeed, in testimony before Japan's House of Representatives, a government official argued that regional market integration could "make it possible to offer consumers lower electricity rates" (House of Representatives, 2014). Without integration, high-cost generators may operate in one region to meet local demand while lower-cost generators elsewhere sit idle, keeping electricity prices unnecessarily high. These inefficiencies were especially costly in our setting, as each region

had to rely more heavily on high-cost generators within its own territory to replace lost nuclear supply and meet local demand. We estimate that regional market integration lowers the average log electricity price by 2.8 log points and reduces price-induced output losses by 32.3% relative to the baseline without integration.

This paper makes several contributions. First, we provide the first micro-level evidence on how adverse shocks to electricity supply affect industrial production in an advanced economy. Decades of research have examined the economic effects of oil and natural gas shocks (Hamilton, 1983; Rotemberg and Woodford, 1996; Kilian, 2008; Hamilton, 2013). Yet oil and natural gas account for less than one-quarter of global electricity generation (IEA, 2026), raising questions about whether these findings generalize to electricity supply shocks. A substantial body of research has also examined electricity supply disruptions, focusing largely on outages in developing economies. (Fisher-Vanden et al., 2015; Allcott et al., 2016; Cole et al., 2018; Fried and Lagakos, 2023).[5] A smaller literature examines how electricity prices affect firm performance in advanced economies (Kahn and Mansur, 2013; Bijnens et al., 2022), including studies exploiting changes in electricity taxes, tariffs, and incentives (Gerster and Lamp, 2024; von Graevenitz and Rottner, 2024).[6] We instead examine large, persistent electricity supply shocks—an increasingly important concern as economies electrify—and document substantial output losses.

Second, we contribute to a longstanding literature on why shocks to inputs with small expenditure shares can cause large output losses. Standard economic theory predicts that small shocks to inputs with small cost shares should have limited effects on output (Solow, 1957; Hulten, 1978). Empirical evidence suggests otherwise: oil and intermediate-input shocks can generate output losses far exceeding these predictions (Rotemberg and Woodford, 1996; Barrot and Sauvagnat, 2016; Boehm et al., 2019). Previous studies propose several amplification mechanisms, including limited input substitution and reduced capital utilization (Atkeson and Kehoe, 1999; Finn, 2000; Atalay, 2017; Baqaee and Farhi, 2019). Yet much of the existing work relies on aggregate data that obscure how individual producers adjust or on calibrated models that do not directly estimate key production parameters. We instead use establishment-level data and find that manufacturers adapted little to the shocks. Ex-

[5]Several studies of blackouts document substantially larger effects than those we estimate. For example, Allcott et al. (2016) estimate that electricity shortages reduce plant revenues in India by 5–10%, roughly three to six times our estimated output decline of 1.7%. For Sub-Saharan Africa, Cole et al. (2018) estimate that reducing outages to South African levels would increase firm sales by 85.1%. Although the shocks and outcomes differ across settings, the smaller effects in our setting may partly reflect a different margin of adjustment: establishments continue to operate at higher marginal costs, whereas blackouts can halt production.

[6]A substantial literature also examines how households respond to electricity prices and conservation policies, including during electricity crises (Reiss and White, 2008; Leighty and Meier, 2011; Ito, 2015; Costa and Gerard, 2021; He and Tanaka, 2023).

ploiting quasi-experimental variation in electricity supply, we further show that limited input substitution and capital idleness amplified output losses far beyond the standard factor-share benchmark.

This paper also contributes to research on nuclear phase-outs and their economic consequences.[7] Major nuclear accidents have repeatedly reshaped energy policy worldwide. After Chernobyl, Italy exited nuclear power; after Fukushima, Germany accelerated its nuclear phase-out, Japan rapidly shut down nuclear generation, and countries such as Switzerland and South Korea adopted or considered policies to reduce their reliance on nuclear power. A growing literature shows that nuclear shutdowns can impose sizable costs by removing low-marginal-cost generation and increasing reliance on fossil fuels. These costs include higher electricity prices (Davis and Hausman, 2016), health damages from increased air pollution (Severnini, 2017; Jarvis et al., 2022), and increased mortality from reduced household electricity consumption (Neidell et al., 2021; He and Tanaka, 2023). We extend this literature by studying a less-examined downstream effect: industrial production. We show that the abrupt shift in nuclear policy harmed industrial production by raising electricity prices.

2 Institutional Background

2.1 Japan's Regulated Electricity Markets

During our study period, Japan's electricity sector was heavily regulated, generating wide regional variation in electricity supply shocks following major disruptions to the electricity system. Japan's mainland has nine regional electricity markets. A single traditional utility dominated each market, supplying more than 90% of local electricity, while wholesale trading was negligible (JEPIC, 2019). In particular, three features of Japan's electricity system contributed to regional variation in these supply shocks.

First, regions differed substantially in their electricity supply mix and reserve margins—two factors that contributed to regional variation in supply shocks. In 2010, nuclear power accounted for 17.4% of total peak-day electricity supply across the nine mainland regions (ESDV, 2012). Nuclear power played a relatively large role partly because Japan imports nearly all the fossil fuels it consumes and the government had long promoted nuclear generation to strengthen energy security. Nuclear reliance nevertheless varied substantially across regions because reactor siting depended on access to cooling water, compliance with seismic

[7] Existing studies document the short- and long-term impacts of nuclear accidents and related disasters. Such events can disrupt supply chains (Carvalho et al., 2021), affect health (Almond et al., 2009; Danzer and Danzer, 2016), alter perceptions of nuclear risk and induce anxiety (Fu et al., 2026), and reduce housing and land prices (Kawaguchi and Yukutake, 2017).

safety requirements, and local political acceptance. The nuclear share ranged from 0% in Chugoku and 9.2% in Chubu to 25.6% in Kansai and 29.1% in Shikoku, as shown in Panel A of Figure 1. Electricity reserve margins also varied across regions. In 2010, Hokkaido and Shikoku had margins of 16.4% and 17.7%, providing larger buffers against the loss of nuclear supply. By contrast, Kansai and Tokyo had margins of only 5.7% and 6.9%, leaving them more exposed to electricity shortages following the nuclear shutdown.

Second, regulation limited interregional transmission and trade, preventing local electricity supply shortfalls from being smoothed across regions. Under the Electricity Business Act, each utility could monopolize electricity supply within its service area but had a legal obligation to maintain reliable local supply (Government of Japan, 1964). Before the accident, most regions traded little electricity with one another (Panel A of Figure 1). During severe electricity supply shocks, exporting electricity became even more difficult because doing so could jeopardize supply to utilities' own customers. Following the Fukushima nuclear accident, interregional electricity transfers indeed remained extremely small, often below 1% of peak-day demand (see Section 4.2).[8]

Third, under Japan's regulatory system, changes in electricity supply costs translated directly into changes in regional electricity prices. Japan used cost-of-service regulation, under which utilities charged customers prices that covered electricity supply costs plus a regulated return. Replacing nuclear power with more expensive thermal generation therefore raised retail prices, especially in regions with greater pre-accident nuclear reliance. Retail electricity prices rose with a delay because of lengthy regulatory approval processes.

2.2 The Fukushima Nuclear Accident and Nuclear Phase-Out

The Fukushima nuclear accident was, in the words of UNSCEAR (2021), "the worst civil nuclear disaster since the one at Chernobyl in 1986." The disaster began with the March 2011 Great East Japan Earthquake—Japan's largest recorded earthquake and the world's fourth-largest since 1900. The earthquake and ensuing tsunami killed more than 15,000 people (U.S. DOI, 2011; USGS, 2011). Tsunami waves 14–15 meters high hit the Fukushima Daiichi nuclear power plant and disabled its cooling systems (IAEA, 2015). As a result, several reactor cores melted down, releasing large amounts of radioactive material. This accident immediately changed public and regulatory attitudes toward nuclear safety.

The accident led to sweeping changes in Japan's electricity policy. It immediately changed public and regulatory attitudes toward nuclear safety and intensified opposition to nuclear

[8] Although physical constraints existed, most notably the incompatibility between eastern Japan's 50 Hz grid and western Japan's 60 Hz grid, the primary barrier to equalizing supply shocks was institutional rather than technological.

power. Japan gradually shut down its nuclear reactors, and all were offline by May 2012. The government subsequently imposed safety standards so stringent that most reactors could not meet them and remained offline for years. Nuclear power thus nearly disappeared from Japan's electricity supply—a shift later described as "a major turning point in Japan's energy policy" (ANRE, 2022). Nuclear power's share of peak supply was just 1.4% in 2013 and remained extremely low at 2.9% in 2018 (METI, 2016; OCCTO, 2018).

The nuclear shutdown triggered a nationwide electricity supply crisis by removing a source that had supplied 17.4% of peak-day electricity in 2010. In mid-2011, the IEA described Japan's situation as "perhaps one of the most severe electricity shortfalls in history" (IEA, 2011).

To address the electricity supply shocks, policymakers adopted both demand- and supply-side measures. On the demand side, the government used moral suasion to maintain electricity-system reliability and prevent widespread blackouts.[9] The government encouraged manufacturers to improve energy efficiency, dim lighting, reduce air-conditioning use, adjust work schedules, and shift operations away from peak hours (Kakuchi, 2011). On the supply side, the government eased regulatory requirements to facilitate additional thermal generation. Utilities reactivated idle fossil-fuel plants, increased output from existing thermal generators, and added new thermal capacity. Replacing nuclear generation with higher-marginal-cost thermal generation raised electricity generation costs, which were largely passed through to consumers as higher retail prices. Regions that relied more heavily on thermal generation to replace lost nuclear output consequently faced larger price increases.

Panels B and C of Figure 1 illustrate the consequences of the Fukushima nuclear accident for Japan's electricity markets. Panel B shows the sharp and persistent decline in nuclear supply. Increased supply from thermal plants only partially offset this loss, leaving total peak-day electricity supply below its pre-accident level. Panel C shows that electricity prices rose through 2014 and remained elevated through 2018, despite a temporary decline. Manufacturers therefore faced persistently higher electricity input costs, with substantial variation across regions.

3 Data

3.1 Peak-Day Electricity Demand and Supply

We assemble regional-level data on peak-day electricity demand and supply from 2005 to 2018 to measure electricity supply shocks. Our primary data source is the Electricity Supply–

[9] In summer 2011, the government mandated a 15% reduction in peak electricity use relative to 2010, but only for large establishments in the Tokyo and Tohoku service areas (METI, 2011).

Demand Verification Subcommittee (ESDV, 2012),[10] a government body that convenes annually to assess summer and winter electricity supply conditions. The subcommittee collects detailed information on regional supply capacity and demand on peak days, which the national government uses to design and implement interventions in electricity markets. In designing its demand-reduction policies, the government used 2010 as the benchmark year and assessed supply adequacy using peak-day rather than peak-hour demand.

3.2 Census of Manufactures

We use plant-level administrative data from the Census of Manufactures (CMF) for 2005–2018, compiled by the Ministry of Economy, Trade, and Industry (METI) (METI, 2012). The census annually surveys all manufacturing establishments with 30 or more employees, covering approximately 80% of total manufacturing output in Japan. Japan's CMF is comparable to manufacturing census data used in studies of the United States and other major economies in scope and structure.

The CMF provides detailed plant-level information on outputs, inputs, and production characteristics. We measure output as revenue adjusted for inventory changes. We construct capital stock as the reported book value of land, buildings, and machinery, net of depreciation. We compute capital rent by multiplying each plant's deflated capital stock by an industry-year rental rate. We measure electricity purchases, materials, and labor using expenditure data, convert all monetary values to a tax-inclusive basis, and deflate them. Because the data do not report plant-level electricity prices, we deflate electricity expenditures using average regional electricity prices. This approach is appropriate in Japan's heavily regulated electricity market, where prices were largely set by regional utilities through tariff schedules (JEPIC, 2019). Each plant is assigned a two- to four-digit industry code.[11] Throughout the paper, we use the three-digit classification as our baseline definition of an "industry" unless otherwise noted.

We construct the analysis sample using three main restrictions. First, we exclude establishments located in municipalities severely affected by the Great East Japan Earthquake and the Fukushima nuclear accident, using an affected-area definition consistent with Carvalho et al. (2021). Second, for each outcome regression, we restrict the sample to establishments observed in both the baseline year, 2010, and the outcome year, so that the dependent variable, the change in log outcomes relative to 2010, is defined. Third, we restrict the CMF sample to establishments with nonmissing and positive values for output and primary input

[10] We supplement these data with information from the Agency for Natural Resources and Energy (ANRE) (ANRE, 2022) and utilities' power supply planning documents.

[11] The data include 24 two-digit industries, 153 three-digit industries, and 546 four-digit industries.

expenditures, including labor, materials, electricity, and capital. We winsorize these output and input variables at the 1st and 99th percentiles to mitigate the influence of outliers.

The CMF includes several additional features useful for our analysis. Firm identifiers allow us to link establishments owned by the same firm and study whether activity shifts across regions within multi-plant firms. A subset of establishments reports product-level information, enabling us to analyze product switching. We construct a product-switching indicator and measure each plant-year's product-mix electricity intensity. Finally, we use publicly available aggregate plant-count data from METI in 2006, 2009, 2014, and 2016 to examine plant turnover. We rely on these aggregate data because our restricted-access microdata include only plants with 30 or more employees. Appendix A-1 provides additional details on the CMF data and variable construction.

3.3 Energy Consumption Statistics

We complement the CMF with plant-level administrative data from the Energy Consumption Statistics (ECS) provided by ANRE ((Agency for Natural Resources and Energy, 2199)). These surveys report physical quantities of fuel inputs, including natural gas (city gas, LPG, and LNG), gasoline, kerosene, heavy oil, diesel oil, and steam. We convert all fuels into a common unit based on their heat content to construct a standardized measure of energy use. A key advantage of the ECS is that it distinguishes fuel used directly in production from fuel used for on-site electricity generation through boilers, cogeneration systems, or diesel generators. Although the ECS is designed to be representative, it places greater emphasis on large and energy-intensive plants, and its sampling frame differs from that of the CMF in industry coverage and timing. Appendix A-2 provides additional details on the surveys' coverage and the construction of the energy-use measures.

3.4 Other datasets

We supplement the plant-level data with additional regional data for robustness checks. To account for supply-chain disruptions from the earthquake and tsunami, we use prefecture-by-industry input-output tables from RIETI (RIETI, 2005). We also collect prefecture-level data on local economic conditions, including gross prefectural product, population, construction investment, consumption, and labor- market conditions (MIC, 2025). See details in Appendix A-3.

3.5 Summary Statistics

Table 1 presents summary statistics. In the CMF sample (N = 437,456 plant-year observations), the average plant generates approximately 4,999 million Japanese yen in annual output (about 50–60 million in 2010 US dollars) and employs 127 workers. Approximately 2.8% of establishments export, and 42% belong to multi-plant firms. On average, labor, capital, and materials account for 22%, 3.3%, and 51% of revenue, respectively, while electricity represents a relatively small share at 2.0%. In the pooled ECS–POFMS sample (N = 380,682 plant-year observations), approximately 13% of observations are classified as generator-owning. On average, plants purchase 20 TJ of electricity from the grid and generate 3.6 TJ on site.

4 Empirical Strategy

4.1 Conceptual Framework

We begin with a conceptual framework that guides our empirical analysis. Manufacturing plants produce output (Y) using capital (K), labor (L), materials (M), and electricity (E), together with Hicks-neutral productivity Ω. The production function is

$$Y = \Omega F(K, L, M, E), \tag{1}$$

where $F(\cdot)$ is differentiable. For this benchmark, suppose plants take output and input prices as given.

Differentiating $\ln Y$ with respect to electricity supply shock severity $\mathcal{S}$ yields

$$\frac{d \ln Y}{d\mathcal{S}} = \theta_E \frac{d \ln E}{d\mathcal{S}} + \theta_M \frac{d \ln M}{d\mathcal{S}} + \theta_L \frac{d \ln L}{d\mathcal{S}} + \theta_K \frac{d \ln K}{d\mathcal{S}} + \frac{d \ln \Omega}{d\mathcal{S}}, \tag{2}$$

where $\theta_X \equiv \partial \ln F(\cdot)/\partial \ln X$ denotes the output elasticity with respect to input $X \in \{K, L, M, E\}$. Appendix B-2 provides the detailed derivation.

Equation (2) shows that electricity supply shocks may affect output through several margins. They may reduce electricity use directly, but they may also affect labor, materials, capital, and productivity. Our empirical analysis therefore estimates the effects of electricity supply shocks on output, electricity use, other inputs, and productivity, except for capital, which is measured with substantial error.

This framework provides a benchmark for the direct output effect of an electricity supply shock. Consider a small, transitory, and unexpected shock, and evaluate the marginal product of electricity at the plant's pre-shock optimum. Holding labor, materials, capital, and

productivity fixed at L^*, M^*, K^*, and Ω^*, respectively, the direct effect is approximated by

$$\left.\frac{d\ln Y}{d\mathcal{S}}\right|_{L=L^*,\, M=M^*,\, K=K^*,\, \Omega=\Omega^*} \approx \theta_E^* \frac{d\ln E}{d\mathcal{S}}. \tag{3}$$

The first-order condition links the electricity output elasticity to the electricity revenue share. Since electricity accounts for only about 2% of revenue on average, this benchmark predicts modest output losses from electricity reductions alone. This helps explain why the existing literature often finds modest, and sometimes null, output impacts of input shocks in electricity markets (von Graevenitz and Rottner, 2024; Wolverton et al., 2022).

This benchmark is not a full prediction of the plant response to the shocks we study. Electricity supply shocks after the Fukushima accident were large, persisted for years, and became increasingly predictable after the near-complete shutdown of nuclear reactors. Over such a horizon, plants can re-optimize materials, labor, and, to some extent, capital.

4.2 Constructing a Measure of Electricity Supply Shocks

We construct a measure of regional electricity supply shocks based on peak-day supply-demand conditions. Peak days are central to blackout risk because electricity supply must meet demand in real time, including on the highest-demand days.

We define shock intensity as the gap between required and available supply in a given region and year, normalized by pre-accident peak demand. Intuitively, it measures the demand reduction—or equivalent supply increase—needed to avoid large-scale blackouts in that region during that year. Specifically, we define the shock as:

$$\mathcal{S}_{rt} = \frac{1}{D^{\text{peak}}_{r,2010}} \Big(\underbrace{1.05\, D^{\text{peak}}_{r,2010}}_{\text{required supply}} - \underbrace{S^{\text{plan,peak}}_{rt}}_{\text{available supply}} \Big) \cdot \mathbb{1}\left\{ S^{\text{plan,peak}}_{rt} < 1.05\, D^{\text{peak}}_{r,2010} \right\}. \tag{4}$$

where $D^{\text{peak}}_{r,2010}$ denotes realized demand on the 2010 peak day in region r, which the government used as the pre-accident benchmark. Required supply equals this pre-accident peak-day demand plus a 5% reserve margin. Reserve margins provide a buffer against demand fluctuations driven by weather and economic activity; we follow the government's 5% reserve-margin threshold for maintaining reliable supply (ESDV, 2012). $S^{\text{plan,peak}}_{rt}$ is planned available supply capacity for the peak day in year t. Available supply declines when existing nuclear capacity is phased out and increases when supply-side measures, such as restarting retired thermal plants or constructing new thermal plants, are taken. The indicator sets the shock to zero in region-years in which available supply exceeds required supply.

Electricity supply shocks exhibit substantial spatial and temporal variation. Panel A of Figure 2 plots the geographic distribution of these shocks, revealing widespread exposure

across Japan. Panel B displays the time series of the average shock measure. Following the 2011 disaster, shock intensity spiked sharply. While shocks persisted throughout the study period, their severity fluctuated significantly across regions and over time.

Appendix Table B1 reports peak-day supply from different energy sources, demand, and interregional transmission volumes for four major regions. Total supply declined relative to 2010 in all regions, largely because of the nuclear phase-out. Transmission was severely limited.[12] When Tokyo faced its most severe supply deficits, it received virtually no electricity from other regions or non-utility suppliers. Instead, Tokyo remained a net exporter, sending roughly 0.1% to 0.5% of its pre-accident capacity to other regions (Panel B). Similarly, even during severe shocks in 2016, Kansai exported power rather than importing it (Panel D).

4.3 Econometric Specification

We estimate the effects of electricity supply shocks on the outcomes in equation (2). We begin with output and estimate the following specification:

$$\Delta \ln Y_{ijrt} = \rho_1 \mathcal{S}_{rt} \mathbb{1}\{2011 \leq t \leq 2012\} + \rho_2 \mathcal{S}_{rt} \mathbb{1}\{2013 \leq t \leq 2018\} + \mu_{jt} + \epsilon_{ijrt}, \quad (5)$$

where $\Delta \ln Y_{ijrt} = \ln Y_{ijrt} - \ln Y_{ijr,2010}$ is the change in plant-level log output relative to 2010, the year before the nuclear accident, for plant i in industry j, region r, and year t. This baseline is consistent with our shock measure, defined relative to 2010 electricity use (ESDV, 2012). The variable $\mathcal{S}_{rt}$ denotes the intensity of the local electricity supply shock, μ_{jt} denotes industry-by-year fixed effects, and ϵ_{ijrt} is the error term.

The coefficients ρ_1 and ρ_2 capture the effects of the shock, $d \ln Y / d\mathcal{S}$, during the phase-out period (2011–2012) and the post-phase-out period (2013–2018), respectively. Conditional on industry-by-year fixed effects, we identify these coefficients from cross-regional variation in shock intensity, net of nationwide industry-specific shocks. We focus primarily on the post-phase-out period because it captures persistent and severe supply disruptions. Immediately after the accident, reactors were gradually suspended and policy uncertainty remained high, so plants may have perceived the disruption as temporary. By 2013, however, nuclear power had been fully phased out, and the resulting electricity supply shocks persisted through 2018 and beyond.

We also estimate event-study-style coefficients to examine pre-trends and dynamic effects. Because electricity supply shocks vary over time, we use $\bar{\mathcal{S}}_r$, the mean region-level shock over 2013–2018, for pre-accident years and the year-specific realized shock, $\mathcal{S}_{r\tau}$, for post-accident

[12] Here, "transmission" refers to electricity imports from other regional utility monopolies or non-utility suppliers (e.g., independent power producers).

years. We estimate

$$\Delta \ln Y_{ijr\tau} = \sum_{t \neq 2010} \beta_t \mathbf{1}\{\tau = t\} \left[\mathbf{1}\{t \leq 2009\}\bar{\mathcal{S}}_r + \mathbf{1}\{t \geq 2011\}\mathcal{S}_{r\tau}\right] + \mu_{jt} + \varepsilon_{ijr\tau}, \tag{6}$$

where τ indexes calendar years in the sample. Statistically insignificant estimates of β_t in pre-accident years support the assumption that high- and low-shock regions followed similar outcome trajectories before the accident. The post-accident coefficients describe how the relationship between realized supply shocks and the outcome evolves over time.

We estimate the effects on electricity use, materials, labor, and productivity analogously by replacing output with each corresponding outcome variable. These estimates correspond to the input and productivity terms in equation (2).

We weight regressions by the log of plant-level total salary in 2010. We conduct inference using the wild cluster bootstrap, with regions as clusters. Because treatment assignment varies across only nine regions, conventional cluster-robust asymptotic approximations may be unreliable and prone to over-rejection (Cameron and Miller, 2015). Following Canay et al. (2021) and Roodman et al. (2019), we use Rademacher weights and report wild-bootstrap p-values and confidence intervals rather than conventional cluster-robust standard errors.

4.4 Instrumenting for Electricity Supply Shocks

Regional electricity supply shocks may be endogenous because of post-accident policy responses. Some regions reactivated dormant thermal plants or restarted nuclear reactors. These supply-side adjustments may be correlated with unobserved local economic conditions, such as recovering industrial demand or political influence.

To address this concern, we use an instrumental-variables approach. Regions that relied more heavily on nuclear power before the accident faced larger electricity supply shocks because nuclear generation was almost completely phased out. At the same time, regions with larger pre-accident reserve margins had greater buffer capacity and were better able to absorb the loss of nuclear supply. We therefore construct the instrument vector, $\mathbf{Z}_{rt}$, from predicted nuclear-supply losses and initial reserve margins:

$$\mathbf{Z}_{rt} = \left(Sh^{\text{nuclear}}_{r,2010} \times g^{\text{nuclear}}_{t,-r},\ Sh^{RM}_{r,2010}\right). \tag{7}$$

Here, $Sh^{\text{nuclear}}_{r,2010}$ denotes the 2010 nuclear-generation share in region r, and $g^{\text{nuclear}}_{t,-r}$ is the leave-one-out national growth rate of nuclear generation. The second instrument, $Sh^{RM}_{r,2010}$, denotes region r's reserve margin in 2010. All variables are normalized by 2010 peak-day demand to match the scale of the shock measure.

4.5 Production Function Estimation

4.5.1 CES Production Function

We estimate a constant-elasticity-of-substitution (CES) production function, assuming that plants are price takers in output and input markets. For plant i in year t, output is given by

$$Y_{it} = \exp(\omega_{it} + \epsilon_{it}) \left(\alpha_K K_{it}^{\rho} + \alpha_L L_{it}^{\rho} + \alpha_M M_{it}^{\rho} + \alpha_E E_{it}^{\rho}\right)^{1/\rho}, \tag{8}$$

where ω_{it} denotes plant-level log productivity and ϵ_{it} is an idiosyncratic ex post production shock. The elasticity of substitution is $\sigma \equiv 1/(1-\rho)$, and the share parameters are positive and normalized such that $\alpha_K + \alpha_L + \alpha_M + \alpha_E = 1$.

Taking logs, equation (8) becomes

$$y_{it} = f(k_{it}, l_{it}, m_{it}, e_{it}) + \omega_{it} + \epsilon_{it}, \tag{9}$$

where lowercase variables denote logs and $f(\cdot)$ is the log of the CES aggregator.

We adopt the following timing assumptions. Plants first choose the quasi-fixed inputs, labor and capital, before observing current-period productivity, ω_{it}. Productivity follows a first-order Markov process,

$$\omega_{it} = h(\omega_{it-1}) + \eta_{it}, \tag{10}$$

where η_{it} satisfies $\mathbb{E}[\eta_{it} \mid k_{it}, l_{it}] = 0$. After observing ω_{it}, plants choose the flexible inputs, electricity and materials, conditional on their predetermined labor and capital. Finally, after all input choices have been made, plants experience the ex post shock ϵ_{it}, which was unanticipated at the time of those choices and satisfies $\mathbb{E}[\epsilon_{it} \mid k_{it}, l_{it}, m_{it}, e_{it}, \omega_{it}] = 0$.

4.5.2 Estimation Strategy

We estimate the structural parameters of the CES production function separately for each of the 153 three-digit manufacturing industries, allowing technology to differ across industries while imposing a common production technology within each industry.

We recover the CES parameters in two steps. First, we estimate industry-specific output elasticities for each input. We identify the output elasticities of the flexible inputs, electricity and materials, using static first-order conditions following Gandhi et al. (2020), and estimate those of the quasi-fixed inputs, labor and capital, using a control-function approach (e.g., Olley and Pakes, 1996; Ackerberg et al., 2015). Second, conditional on a candidate elasticity of substitution (σ), we recover the CES distribution parameters (α) from the estimated output elasticities. Under cost minimization, an input's normalized output elasticity equals its share of total input cost. The CES first-order conditions then map these cost shares into

the distribution parameters.

This approach offers three advantages. First, it allows us to estimate a gross-output production function using static first-order conditions. As Gandhi et al. (2020) show, standard proxy-variable approaches generally cannot identify flexible-input output elasticities in gross-output production functions without additional variation in flexible-input demand, which is often unavailable. Second, the first step delivers output elasticities, so the implied CES share parameters can be recovered for alternative values of σ without re-estimating the output elasticities. Third, estimating the model separately by three-digit industry accommodates technological heterogeneity across manufacturing industries. This heterogeneity is central to the counterfactual analysis because it allows us to quantify the separate contributions of physical rationing and higher electricity prices to production losses. Without technological heterogeneity, the two channels would generate identical production responses.

4.5.3 Estimating Output Elasticities and Productivity

Flexible Inputs In the first stage, we estimate output elasticities of the flexible inputs: $\theta_E \equiv \partial f/\partial e$ (electricity) and $\theta_M \equiv \partial f/\partial m$ (materials). Under the assumption that plants are price takers in both input and output markets, the first-order condition for a flexible input implies that its log revenue share equals its log output elasticity minus a mean-zero idiosyncratic shock. For electricity,

$$\ln s_{E,it} = \ln \theta_E - \epsilon_{it}, \tag{11}$$

where $s_{E,it} \equiv \frac{P_t^E E_{it}}{P_{it} Y_{it}}$ and $P_{it}Y_{it}$ is nominal revenue. Equation (11) implies that θ_E is identified by the conditional mean of the log electricity expenditure share.[13] We obtain θ_M analogously using the materials share.

Quasi-fixed Inputs In the second stage, we estimate the output elasticities of the quasi-fixed inputs, labor and capital. Using the first-stage estimates of the flexible-input output elasticities, $\widehat{\theta}_E$ and $\widehat{\theta}_M$, we construct the following partial residual for each plant-year observation:

$$\widehat{\mathcal{Y}}_{it} \equiv y_{it} - \widehat{\theta}_E e_{it} - \widehat{\theta}_M m_{it}. \tag{12}$$

This partial residual represents log output net of the estimated contributions of electricity and materials. For candidate values of (θ_L, θ_K), we define implied productivity as

$$\widehat{\omega}_{it}(\theta_L, \theta_K) \equiv \widehat{\mathcal{Y}}_{it} - \theta_L l_{it} - \theta_K k_{it} - \widehat{\epsilon}_{it}. \tag{13}$$

[13] We impose the normalization $\mathbb{E}[\exp(\epsilon_{it}) \mid k_{it}, l_{it}, m_{it}, e_{it}] = 1$, so that $\mathbb{E}[\ln s_{E,it} \mid k_{it}, l_{it}, m_{it}, e_{it}] = \ln \theta_E$.

Here, $\widehat{\epsilon}_{it}$ is obtained in the first stage.[14] For each candidate pair (θ_L, θ_K), we use the implied productivity series to estimate the conditional expectation function $h(\cdot)$ and construct the corresponding productivity innovation as $\widehat{\eta}_{it} = \widehat{\omega}_{it} - \widehat{h}(\widehat{\omega}_{i,t-1})$. We then estimate θ_L and θ_K by setting the sample analogues of $\mathbb{E}[\eta_{it} l_{it}] = 0$ and $\mathbb{E}[\eta_{it} k_{it}] = 0$ equal to zero. Because there are two parameters and two moment conditions, the GMM system is exactly identified.

Total Factor Productivity Following Gandhi et al. (2020), we define plant-level log productivity as $\widehat{\text{tfp}}_{it} = \widehat{\omega}_{it} + \widehat{\epsilon}_{it}$, where $\widehat{\omega}_{it}$ is the persistent component and $\widehat{\epsilon}_{it}$ is the unanticipated component of productivity.

4.5.4 Recovering CES Distribution Parameters

For each industry, we first normalize the estimated output elasticities to obtain input cost shares:

$$\hat{s}_j = \frac{\hat{\theta}_j}{\sum_{k \in \{K,L,M,E\}} \hat{\theta}_k}, \qquad j \in \{K, L, M, E\}. \tag{14}$$

Here, $\hat{\theta}_j$ denotes the estimated output elasticity of input j.

Under cost minimization, these normalized output elasticities correspond to input cost shares. Conditional on a candidate elasticity of substitution σ, we recover the CES distribution parameters as

$$\hat{\alpha}_j(\sigma) = \frac{\hat{s}_j^{1/\sigma}}{\sum_{k \in \{K,L,M,E\}} \hat{s}_k^{1/\sigma}}, \qquad j \in \{K, L, M, E\}. \tag{15}$$

This parameterization satisfies $\sum_j \hat{\alpha}_j(\sigma) = 1$ and reproduces the estimated cost shares. Appendix B-3.2 provides the detailed derivation.

Taking the ratio of the recovered distribution parameters for any two inputs j and k gives

$$\frac{\hat{\alpha}_j(\sigma)}{\hat{\alpha}_k(\sigma)} = \left(\frac{\hat{s}_j}{\hat{s}_k}\right)^{1/\sigma}. \tag{16}$$

Thus, inputs with larger estimated cost shares receive larger distribution parameters. Holding the estimated cost shares fixed, a smaller σ implies a more dispersed distribution of these parameters. When substitutability is limited, cost shares are less responsive to differences in the distribution parameters, so larger parameter differences are required to reproduce the ob-

[14] We construct separate estimates of $\widehat{\epsilon}_{it}$ from the electricity and materials share equations and combine them using a weighted average, with weights proportional to the corresponding input expenditure shares in revenue.

served cost-share differences. Conversely, when σ is large and inputs are more substitutable, smaller parameter differences are sufficient.

4.5.5 Production Function Results

Panel A of Table 2 reports output elasticities estimated from the 2005–2010 pre-accident data. Columns (1)–(3) apply the method of Gandhi et al. (2020). Across industries, the estimated elasticities average 0.446 for materials, 0.015 for electricity, 0.354 for labor, and 0.139 for capital. The electricity elasticity varies substantially across industries, from 0.008 at the 25th percentile to 0.017 at the 75th percentile.

Our results broadly align with those of Gandhi et al. (2020) in two respects. First, OLS yields larger elasticities for flexible inputs. Column (4) of Table 2 reports an OLS electricity elasticity of 0.068, more than four times our preferred estimate of 0.015, and a materials elasticity of 0.492, compared with 0.446. These differences are consistent with upward simultaneity bias. Plants increase flexible inputs in response to positive productivity that OLS does not observe, leading OLS to attribute part of the resulting output increase to those inputs. For quasi-fixed inputs, the pattern is mixed: OLS understates the capital elasticity (0.042 versus 0.139) but overstates the labor elasticity (0.423 versus 0.354). Because these biases depend on the joint relationships among input choices and unobserved productivity, we remain agnostic about their precise sources.

Second, our preferred estimator implies greater TFP dispersion than OLS. We calculate the 75/25 and 90/10 ratios within each industry, and report their means across industries in Panel B of Table 2. The mean 90/10 ratio is 2.35 under our preferred estimator and 1.95 under OLS. The corresponding 75/25 ratios are 1.51 and 1.38. This pattern is consistent with transmission bias. Because flexible-input use is positively correlated with productivity, OLS attributes part of the productivity-related variation in output to flexible inputs. Consequently, it overstates their output elasticities and understates productivity dispersion.

5 Electricity Supply Shocks and Plant Outcomes

5.1 Trends in Inputs and Outputs of Manufacturing Plants

We begin by describing aggregate trends in manufacturing inputs, output, and productivity among plants in high- and low-shock regions. We classify regions as high-shock if their mean shock intensity over 2013–2018 is at or above the sample mean of 7.4%. Figure 3 plots our outcome variables over 2005–2018 as log changes relative to their 2010 levels.

Figure 3 shows that, following the 2011 accident, manufacturing inputs and output declined in high-shock regions relative to low-shock regions. In Panel A, electricity use followed

similar trends in the two groups before 2011. After the accident, however, electricity use fell substantially in high-shock regions but changed only modestly in low-shock regions. This divergence widened as nuclear reactors were gradually shut down during 2011–2012 and persisted throughout the complete-shutdown period from 2013 to 2018. Panels B–D show similar, though smaller, divergences in employment, total salary, and material inputs. By contrast, Panel E shows no comparable divergence in productivity. Panel F shows a persistent relative decline in output in high-shock regions after 2011, consistent with the corresponding trends in labor and material inputs, which together account for a large share of production costs.

Regions exposed to more intense shocks experienced larger declines in electricity use, particularly following the complete shutdown. In Figure C1, we first residualize plant-level log changes relative to 2010 levels using industry-by-year fixed effects and then aggregate the residuals across the nine regions. Panel A shows a negative relationship between supply-shock intensity and residualized changes in electricity use. This relationship is stronger during the complete-shutdown period (2013–2018) than during the phase-out period (2011–2012). By contrast, Panel B shows no relationship between subsequent regional supply shocks and pre-accident changes in electricity use. Panel C shows a similar pattern for output: larger shocks are associated with greater declines in output during the complete-shutdown period, but not during the phase-out period. Likewise, Panel D shows no relationship between subsequent regional supply shocks and pre-accident changes in output.

5.2 First-stage Results: Testing the Instruments

We assess instrument relevance by estimating the first-stage relationship between regional electricity supply shocks and our two instruments: predicted changes in nuclear generation and the initial reserve margin. Using region-by-year data from 2011 to 2018, we regress local shock intensity on both instruments, controlling for year fixed effects.

Predicted losses in nuclear generation intensify local electricity supply shocks. Table 3 reports an estimated coefficient of -0.26 on predicted nuclear supply growth during the post-phase-out period (Column (2)). In the average region, nuclear generation accounted for 18% of 2010 peak-day demand, 97% of which was predicted to be lost during 2013–2018. The estimated coefficient therefore implies that these losses increased the local supply shock by 4.4 percentage points ($18\% \times 97\% \times 0.26$). In the most exposed regions, where the predicted loss reached 30% of 2010 peak-day demand, the implied increase was approximately 8 percentage points.

The initial reserve margin mitigates local electricity supply shocks. Table 3 reports an estimated coefficient of -0.82 (Column (2)), consistent with our conjecture that pre-existing capacity buffers helped regions absorb these shocks. The initial peak-day reserve

margin averaged 10%. A region with a reserve margin of only 5%—a critical threshold for maintaining grid reliability—would therefore experience a 4.1 percentage point larger local supply shock than a region at the mean, because its smaller capacity buffer leaves it more exposed to supply shortfalls.

The first stage is strong despite the fact that we have only nine independent clusters. The first-stage F-statistic is 30 for the phase-out period (2011–2012) and 26 for the post-phase-out period (2013–2018).[15]

5.3 The Effects of Supply Shocks on Manufacturing Outcomes

5.3.1 Impacts on Electricity Inputs

This section examines how electricity supply shocks affect manufacturing inputs, productivity, and output. Figure 4 presents event-study estimates based on Equation (6). We regress changes in log outcomes relative to 2010 on local electricity supply shocks. We report both OLS and 2SLS estimates. The 2SLS specification instruments local supply shocks with predicted nuclear-generation losses and initial reserve margins. The figure displays the OLS estimates as blue circles and the 2SLS estimates as red diamonds, together with 90% confidence intervals.

Panel A of Figure 4 shows that electricity supply shocks reduced electricity use in manufacturing. Before the accident, the estimated effects are statistically indistinguishable from zero. This pattern suggests similar pre-trends in regions with high and low supply shocks. Following the accident, electricity use declined, and the effect grew over time. The effect was small in 2011, when firms may have perceived the shock as temporary. It increased in 2012 and persisted through 2018. The OLS and 2SLS estimates are similar in magnitude and statistical precision.

Following the complete shutdown of nuclear power, our estimates imply that manufacturing plants reduced their electricity use by an average of 6.6% during 2013–2018. Table 4 reports the OLS and 2SLS estimates of Equation (5). During the phase-out period (2011–2012), the 2SLS estimate in Panel B implies that a 1 percentage point increase in shock intensity reduced electricity use by approximately 0.21%. The magnitude of this effect increased to 0.89% during the post-phase-out period (2013–2018). Given the mean shock intensity of 7.4% during this period, the estimate implies an average decline in manufacturing electricity use of approximately 6.6%. In a high-shock region such as Tokyo, where mean shock intensity reached 10.8%, the implied decline was approximately 9.6%.

[15]These values exceed the conventional threshold of $F > 10$ suggested by Stock and Yogo (2003). Lee et al. (2022) argue that substantially larger F-statistics may be desirable in some settings to ensure robust instrument relevance.

5.3.2 Impacts on Labor and Materials

Figure 4 shows that electricity supply shocks reduce employment (Panel B), total salary (Panel C), and materials (Panel D), and that these effects grow over time. We show the event-study results in equation (6) with OLS and IV estimates using these outcomes analogously. Across these outcomes, we do not find systematic differences in pre-trends and the OLS and IV estimates are similar in magnitude.

On average, electricity supply shocks reduced employment, total salary payments, and material inputs by 1.3% during the post-phase-out period (2013–2018). Columns (2)–(4) of Table 4 report the estimated effects on these outcomes. The IV estimates in Panel B imply that a 1 ppt increase in the supply shock reduced each outcome by 0.17%. Evaluated at the mean supply shock of 7.4%, these estimates imply an average decline of approximately 1.3% in each outcome.

We further examine effects across worker types and find that non-regular workers experienced disproportionate employment losses. Appendix Table C1 shows little effect on wages per worker. The estimated coefficient for the post-phase-out period is -0.01 (Panel B, Column (1)), suggesting that firms adjusted employment rather than wages. The estimated employment effects are similar for male and female workers: -0.17 in Column (2) and -0.20 in Column (3), respectively. However, the effects differ substantially between regular and non-regular workers. The coefficient for regular workers is small and statistically insignificant at -0.07 (Panel B, Column (4)). By contrast, the coefficient for non-regular workers is -0.90 (Panel B, Column (5)), more than ten times as large in magnitude. These results indicate that firms adjusted employment primarily by reducing non-regular positions. Although non-regular workers accounted for only 10% of manufacturing employment, they accounted for nearly 60% of the implied employment decline.

A back-of-the-envelope calculation suggests that electricity supply shocks reduced manufacturing employment by approximately 97,000 workers per year during 2013–2018. In 2010, manufacturing plants employed 7.7 million workers. Applying our estimated average post-phase-out effect of 1.3% to this baseline yields this figure.[16]

5.3.3 Impacts on Productivity

We estimate the effects of electricity supply shocks on TFP. Theory offers competing predictions. TFP may remain unchanged under a neoclassical model in which productivity is exogenous. Alternatively, it may decline if electricity shocks distort the input mix or lead to

[16]The employment total covers manufacturing plants with at least four employees (METI, 2012). We assume that the post-phase-out estimate applies to aggregate manufacturing employment in 2010 and use the unrounded coefficient to calculate the employment effect.

underutilization of quasi-fixed inputs, consistent with evidence from past oil shocks (Kilian, 2008). Productivity may also rise if unrealized efficiency gains exist and shocks induce manufacturing plants to improve efficiency by adopting electricity-saving technologies or better managerial practices. Anecdotal evidence suggests that Japanese manufacturers undertook extensive energy-conservation efforts, potentially resulting in efficiency gains (Kimura and Nishio, 2016). We measure log TFP as $\widehat{\omega}_{it} + \widehat{\epsilon}_{it}$, using the industry-specific input elasticities estimated from the 2005–2010 data, and estimate the total effect of electricity supply shocks on this measure.

Panel E in Figure 4 provides no evidence that productivity improved in response to the shocks. The point estimates are generally negative after the accident but statistically insignificant throughout the post-accident period. Column (5) of Table 4 reports OLS estimates of -0.03 in both the phase-out and post-phase-out periods. The corresponding 2SLS estimates are -0.04 and -0.07, respectively. Thus, although the point estimates suggest modest productivity losses, they are too imprecise to establish an effect on TFP.

Appendix C-2 examines whether electricity supply shocks induced electricity-biased technical change. Because the electricity–labor ratio depends on both relative factor prices and relative productivity, we remove the variation explained by relative prices to construct a descriptive measure of technical change. This measure evolves similarly in high- and low-shock regions, providing little evidence that more severe supply shocks induced electricity-biased technical change.

5.3.4 Impacts on Output

Electricity supply shocks significantly reduce output, with effects that grow over time through 2015 and then persist thereafter, as shown in Panel F of Figure 4. As with the labor and materials results, we find no significant pre-trends, and the OLS and IV estimates are similar.

The output losses persisted through 2018, seven years after the accident, as shown in Panel F of Figure 4. By September 2013, all nuclear reactors in Japan had been taken offline, while the timing and extent of future restarts remained highly uncertain. Manufacturing firms therefore had reason to view the electricity supply shock as persistent rather than transitory. Unlike the small, transitory-shock benchmark discussed in Section 4.1, manufacturing plants adjusted other inputs, reducing their use of materials and labor and thereby contributing to lower output. The persistence of these losses suggests that plants had limited capacity to adapt to the prolonged disruption in electricity supply.

Electricity supply shocks reduced manufacturing output by an average of 1.7%. Column (6) of Table 4 reports an IV estimate of -0.23 for the post-phase-out period (Panel B), whereas the estimate for the phase-out period is small and statistically insignificant. Evaluated at the mean supply shock of 7.4%, the post-phase-out estimate implies an average

decline in manufacturing output of approximately 1.7%.

The estimated output effect implies a substantial aggregate economic loss. Applying the estimated 1.7% post-shutdown decline to Japan's 2010 manufacturing output of 289 trillion Japanese yen (METI, 2012) yields an annual loss of approximately 4.9 trillion Japanese yen, equivalent to roughly 40–60 billion US dollars depending on the exchange rate. For comparison, this loss corresponds to nearly three years of pre-crisis manufacturing growth: according to the CMF, real manufacturing output grew by an average of approximately 0.6% per year between 1995 and 2007 (METI, 2012).

We benchmark our estimates against previous studies of electricity and broader energy-market shocks. Our estimated decline in manufacturing output is broadly comparable in magnitude to the effects of historical oil shocks. A large literature links major energy-market disruptions to contractions in economic activity, while emphasizing that the magnitude depends on the source of the shock and the empirical specification (Kilian, 2008; Hamilton, 2013). Kilian and Vigfusson (2017) estimate that oil-price shocks associated with the Iranian Revolution and Iraq's invasion of Kuwait reduced U.S. real GDP by 0.5–3.5% over the following two years, depending on the model specification. Recent estimates also suggest that disruptions to Russian energy supplies reduced German GDP by 0.5–3% in the short run (Bachmann et al., 2024).

In contrast, studies of relatively modest and temporary changes in electricity prices generally find small or statistically insignificant effects on output and competitiveness (Wolverton et al., 2022; von Graevenitz and Rottner, 2024; Gerster and Lamp, 2024). As discussed in Section 4.1, one explanation is that plants do not reoptimize other inputs in response to modest, temporary price changes, limiting the resulting effects on output. These contrasting findings are consistent with our prediction that the scale and persistence of an energy shock shape its effects on output.

Evidence from developing countries shows larger output losses from grayouts and blackouts (Fisher-Vanden et al., 2015; Allcott et al., 2016; Cole et al., 2018; Fried and Lagakos, 2023). For example, Allcott et al. (2016) estimate that these disruptions reduce Indian manufacturers' annual revenues by 5–10%, while Cole et al. (2018) estimate that reducing power outages to South African levels would increase firm sales by 85.1%. Although measured differently, these effects exceed ours, plausibly because grayouts constrain production and blackouts halt it entirely during outages. Japanese manufacturers generally remained in operation but faced persistently higher electricity prices and limited peak-period rationing.

Other disruptions provide additional benchmarks. Natural disasters reduce sales growth at connected firms by approximately 2–4 percentage points through input–output linkages (Barrot and Sauvagnat, 2016; Carvalho et al., 2021). These effects arise immediately after individual disasters, whereas our estimated losses persist over six years. Environmental

regulation can also produce sizable losses: Greenstone (2002) estimates that nonattainment regulation reduced shipments among affected high-emitting industries by 3.6–7.2% over five-year intervals.[17] Although the outcomes and time horizons differ, these estimates indicate that our annual output loss of 1.7% is economically substantial. Moreover, unlike localized natural disasters and place-based environmental regulations, our shocks were nationwide and exposed virtually all manufacturing industries to adverse electricity-supply conditions.

5.4 Heterogeneous Impacts

The heterogeneity analysis sheds light on the channels through which electricity supply shocks affect manufacturing outcomes. We examine whether the effects vary with plant- and industry-level electricity and capital intensity, measured in 2010.[18] For each measure, we split the sample at the median, estimate Equation (5) separately for the two groups, and report the 2SLS estimates for the complete-shutdown period.

The effect of electricity supply shocks on electricity use varies little with baseline electricity or capital intensity, measured at either the plant or industry level. Panel A of Figure 5 shows that the subgroup estimates generally lie close to the full-sample estimate of -0.89. For each sample split, we fail to reject equality between the estimates for the high- and low-intensity groups. This limited heterogeneity is consistent with Japanese policy: plants within the same region generally faced common rationing requirements and electricity price increases, regardless of their characteristics.

By contrast, the output impacts vary sharply with capital intensity. Panel B of Figure 5 reports a coefficient of -0.25 for capital-intensive plants, compared with 0.00 for plants with low capital intensity. The corresponding coefficient is -0.38 for plants in capital-intensive industries, compared with 0.05 for those in less capital-intensive industries. We find no comparable heterogeneity by electricity intensity.

Electricity use declines similarly across subgroups, yet output falls more at capital-intensive plants and in capital-intensive industries. This contrast is consistent with electricity supply shocks amplifying output losses through capital utilization rather than through larger reductions in electricity use. The energy-shock literature emphasizes this mechanism: electricity shortages can reduce the operating hours of installed machinery, lowering effective capital services and output (Berndt and Wood, 1975; Finn, 2000). Motivated by this evidence, Section 7.3.1 incorporates electricity-dependent capital utilization into the production technology.

[17] Across all regulated pollutants, the estimated cumulative loss in shipments totaled approximately 75 billion US dollars in nonattainment counties between 1972 and 1987, measured in 1987 US dollars.

[18] Electricity intensity is electricity expenditure divided by revenue, and capital intensity is the capital stock divided by revenue.

5.5 Alternative Explanations and Robustness Checks

5.5.1 Supply-Chain Disruptions and Local Demand Shocks

We examine whether supply-chain disruptions provide an alternative explanation for our finding that electricity supply shocks led to large output contractions. Previous research documents that damage from the Great East Japan Earthquake propagated through input–output linkages (Carvalho et al., 2021). Such disruptions could therefore have contributed to the estimated output declines.

We assess this alternative explanation by controlling for two measures of exposure to supply-chain disruptions (see Appendix A-3 for details on their data and construction). The first captures direct exposure as the share of intermediate inputs sourced from the three most severely affected prefectures: Iwate, Miyagi, and Fukushima. The second captures indirect exposure through upstream linkages, calculated using the Leontief inverse. We interact these pre-disaster, time-invariant exposure measures with year dummies, allowing the effects of supply-chain disruptions to vary flexibly over time.

Table C3 shows that controlling for supply-chain disruptions does not materially alter our estimates. Column (1) reproduces the main 2SLS coefficients during the post-phase-out period: -0.89 for electricity use and -0.23 for output. Controlling separately for direct or indirect exposure yields an electricity coefficient of -0.87 and an output coefficient of -0.20 in both specifications (Columns (2) and (3)). Including both exposure measures produces estimates of -0.88 for electricity use and -0.20 for output (Column (4)). Thus, the supply-chain specifications leave the electricity response essentially unchanged and produce economically meaningful output effects of similar magnitude.

We further investigate whether the observed contractions reflect localized demand shocks rather than supply-side constraints. More-exposed regions may also have experienced greater earthquake damage or weaker consumption, investment, and population growth. We therefore control for time-varying prefecture-level economic conditions, including log gross prefectural product in non-manufacturing sectors, total and working-age populations, construction investment, aggregate consumption, and the job-openings-to-applicants ratio. Adding these controls yields 2SLS coefficients of -0.75 for electricity use and -0.23 for output (Column (5)). Thus, the electricity response is somewhat attenuated, whereas the output point estimate is nearly identical to the baseline, though less precisely estimated.

5.5.2 Robustness Check

Our results are robust to plant-level controls, alternative specifications, and weighting schemes. Column (1) of Appendix Table C4 adds controls for pre-shock (2010) log revenue, electricity, capital, and labor cost shares, export intensity, and multi-plant firm status. The estimated

effects remain stable after including these controls. Column (2) replaces the first-difference specification with a two-way fixed effects specification that includes plant and year fixed effects. Column (3) reports unweighted estimates, which are similar to those from our baseline specification weighted by log total salary.

6 Plants' Responses to Electricity Supply Shocks

6.1 Impacts on Observed Plant Behaviors

Why did electricity supply shocks lead to such large output contractions? To help answer this question, we examine how and to what extent manufacturing plants adapted to these shocks. If plants had limited capacity to adapt, this could partly explain the large observed impacts.

We begin by testing whether plants switched products or increased outsourcing. A natural hypothesis is that plants shifted toward less electricity-intensive products (von Graevenitz et al., 2025) or increased outsourcing to reduce their reliance on electricity (Fisher-Vanden et al., 2015). We use measures of product switching, product-level electricity intensity, outsourcing inputs, and the outsourcing share of material inputs to investigate these adjustment margins (see Appendix A-1.4 for details on data construction). We regress these outcomes on electricity supply shock intensity using both OLS (Panel A) and IV (Panel B), following the specification in equation (5).

We find no evidence that electricity supply shocks induced product switching or increased outsourcing. Table 5 shows that the IV estimates for the post-phase-out period (2013–2018) are small and statistically insignificant, with coefficients of 0.07 for product switching (Column (1)) and 0.03 for product-level electricity intensity (Column (2)), suggesting limited adjustment along the product-choice margin. Columns (3) and (4) likewise provide no evidence that the shocks increased either outsourcing inputs or the outsourcing share of material inputs. If anything, the point estimates are negative.

We further find no evidence that multi-plant firms reallocated activity across establishments, a margin of adjustment to localized input shocks suggested by previous studies (Chen et al., 2025). In our data, 42% of plants belong to multi-plant firms, accounting for 66% of output. Manufacturing firms may respond to localized shocks by shifting production from plants in high-shock regions to plants in low-shock regions, thereby partially maintaining firm-level production. To test this margin, we aggregate the data to the firm-by-region level and examine whether shocks in one region increase the same firm's activity in other regions. For example, shocks in Tokyo may induce the same firm to increase production in Osaka and Hokkaido (see Appendix A-1.4 for details on data construction). Appendix Table D1

provides limited evidence of such reshuffling. In the IV estimates for the post-phase-out period, shocks are associated with an increase in electricity use at the firm's other plants, with a coefficient of 0.84 (Panel B, Column (1)). However, we detect no corresponding response in output, with a coefficient of −0.04 (Column (2)). This absence of reshuffling in output, despite some increase in electricity use at other plants, suggests that firms faced substantial difficulty shifting production across plants.

Additionally, Appendix Table D2 shows no statistically significant effects of supply shocks on plant turnover. In the matched pre-accident sample, approximately 1.2% and 5.5% of manufacturing plants entered and exited the sector, respectively. The IV estimates for entry and exit are −0.10 and 0.18, respectively. At the mean shock of 7.4%, these point estimates correspond to approximately 0.1 fewer entrants and 0.7 additional exits per 1,000 plants ($1.17\% \times (-0.10) \times 7.4\% \times 1{,}000 \approx -0.1$ for entry and $5.46\% \times 0.18 \times 7.4\% \times 1{,}000 \approx 0.7$ for exit). These results suggest limited responses in plant turnover.

6.2 Substitution Between Electricity and Other Fuels

We examine whether plants substituted away from grid electricity and toward alternative energy sources, as electricity and other types of energy are often considered close substitutes.

In theory, such substitution can occur through two channels. First, manufacturing plants may switch to other fuels in production. Some electricity-using processes, such as those involving motor-driven machinery or electric furnaces, may in principle be replaced by fuel-using processes involving boilers, furnaces, or other combustion-based equipment, although such substitution may be costly. Second, plants may substitute indirectly by using alternative fuels for on-site electricity generation. Existing studies document increases in on-site generation in response to higher electricity prices (von Graevenitz and Rottner, 2024) and power outages (Allcott et al., 2016). To examine these two channels separately, we construct two outcome variables: fuel used directly in production and fuel used for self-generation. We then estimate our main specification in equation (5) using each variable as an outcome, applying both OLS and IV.

We find little evidence of substitution away from electricity toward fuel used in production. If these inputs were strong substitutes, electricity supply shocks should cause fuel use to decline less than other inputs, or even to increase. However, we find no such pattern. In Column (1) of Table 6, the post-phase- out IV estimate is −0.35 and statistically imprecise. This estimate is more negative than the corresponding estimates for labor and materials, both −0.17, which are generally considered complements to electricity (Section 5.3). These results provide no evidence that plants substituted fuels used directly in production for electricity.

In contrast, Columns (2) and (3) of Table 6 show that supply shocks increased on-site

self-generation. In 2010, 8.4% owned self-generators. Column (2) shows that the post-phase-out 2SLS estimate for fuel used for on-site generation is 4.23, implying about a 31% increase for a region exposed to the mean shock (4.23×0.074). The corresponding estimate for self-generated electricity is 1.84 (Column (3)), implying an increase of approximately 14% (1.84×0.074).

Our heterogeneity results suggest that on-site generation could mitigate the adverse effects of electricity supply shocks. To examine this possibility, we estimate the main specification in equation (5) separately for plants with and without generators. Electricity supply shocks reduce electricity use—measured in the CMF as electricity purchased from utilities—as shown in Appendix Figure D1. The post-phase-out IV coefficients are -1.11 for plants with generators and -0.85 for plants without generators, implying reductions of approximately 8.2% and 6.3%, respectively, at the mean shock. Despite the decline in grid electricity purchases, output did not fall detectably among plants with generators: the corresponding coefficient is -0.09 and is statistically insignificant. By contrast, output declined among plants without generators, with a coefficient of -0.26. This pattern is consistent with on-site generation helping plants offset reductions in grid electricity purchases and thereby mitigate the adverse effects on output.

We also find an extensive-margin response in generator ownership. Column (4) of Table 6 reports a post-phase-out IV estimate of 0.30, which is statistically significant at the 1% level. However, the effect is modest in absolute terms. For a region exposed to the mean shock, the estimate implies a 2.2-percentage-point increase in ownership ($0.30 \times 0.074 \times 100$), equivalent to approximately 22 additional plants per 1,000 plants. Thus, although the shock induced some adoption, nearly 90% of manufacturing plants remained without on-site generation.

Overall, this section highlights the difficulty manufacturing plants faced in adapting to large-scale and persistent electricity supply shocks. Even using data through 2018, by which time the shocks were broadly expected to persist, we find limited evidence of adjustment through product switching, outsourcing, production reshuffling, plant entry and exit, or direct substitution toward other fuels. Self-generation is an important exception: plants with pre-existing generators increased on-site generation and experienced no detectable output contraction, while some other plants began reporting positive self-generation. Nevertheless, nearly 90% of manufacturing plants remained unable to offset the supply shocks through this channel. This limited scope for adjustment could partly explain the large output losses caused by the shocks.

7 From Reduced-Form Effects to Production Mechanisms

This section uses the production function estimated in Section 4.5 to interpret the reduced-form effects of electricity supply shocks. We predict the impacts of electricity supply shocks using this production function and compare these predictions with the reduced-form estimates. This comparison sheds light on the mechanisms that could reproduce the reduced-form responses

7.1 Simulation Framework

We use the production function to predict electricity use, materials, labor, and output with and without the observed electricity supply shocks. The resulting differences correspond to the reduced-form effects estimated using Equation (2). We use the same sample as in the reduced-form analysis. We use 2010 as the baseline year and average conditions over 2013–2018 to represent the post-phase-out period.

Each plant i is characterized by its 2010 productivity, $\omega_{i,2010}$; output, $Y_{i,2010}$; and generator-ownership status, $G_{i,2010}$. For each industry j, we use the estimated output elasticities to construct the baseline cost shares s_{hj} for $h \in K, L, M, E$. For each candidate elasticity of substitution, σ, we hold these cost shares fixed and recover the corresponding CES distribution parameters, α_{hj}, as described in Section 4.5.4.

We assume that each plant faces isoelastic demand,

$$Y_i = B_i P_i^{-\eta}, \tag{17}$$

where B_i is a plant-specific demand shifter and $\eta > 1$ is the absolute value of the price elasticity of demand. We calibrate B_i so that demand at the baseline price reproduces the plant's observed output in 2010. We set $\eta = 5$, which lies within the range of sector-specific elasticities of substitution estimated for U.S. manufacturing by Shapiro and Walker (2018).

Electricity supply shocks affected manufacturing plants through two channels: quantity rationing and increases in the electricity price, p^E. To model the rationing channel, we follow He and Tanaka (2023) and use the government's regional electricity-saving targets, denoted by $\overline{R}_r$. We impose each region's target on its manufacturing plants during the 90-day shock season, consistent with the government's demand-reduction strategy (ESDV, 2012). The plant-weighted target averages 7.7% in our sample. To model the price channel, we use only the price changes attributable to local supply shocks because observed electricity prices also reflect global energy prices, which contributed to the price increase in 2014 and the declines in 2015–2016. Appendix E-2 shows that a 1-percentage-point increase in shock intensity

raises the log electricity price by approximately 1.12%. At the mean shock intensity, this estimate implies that the average plant faced an electricity price increase of approximately 8.4% during the post-phase-out period.

For each plant, we use its production technology and demand to predict profit-maximizing input use and output with and without the supply shocks. We hold technology and demand fixed across the two scenarios and vary only electricity prices and the rationing constraint. Because rationing applies only during the peak season, we solve the plant's optimization problem separately for the peak and off-peak seasons and aggregate the predictions to obtain annual outcomes. Table E1 summarizes the simulation inputs, and Appendix E-3 describes the procedure.

The simulation also allows responses to differ by access to self-generation. In Section 6.2, we show that plants with generators reduce their purchases from grid utilities but exhibit no meaningful declines in other inputs or output. Based on this evidence, we assume that these plants offset reductions in grid purchases through self-generation and maintain their baseline levels of non-electricity inputs and output.

7.2 Cobb-Douglas Technology

We first predict the impacts of electricity supply shocks assuming $\sigma = 1$, that is, a standard Cobb–Douglas benchmark. We use the model to predict the effects on electricity use, labor, materials, and output and compare these predictions with the reduced-form effects evaluated at the mean shock intensity in Section 5.3.

The Cobb–Douglas benchmark fails to reproduce the reduced-form impacts in two respects. First, it substantially understates the effect of electricity supply shocks on output. As shown in Figure 6, the model predicts an output decline of only 0.6% (Panel B), compared with the reduced-form estimate of 1.7% (Panel A). Second, it misrepresents the relative responses of production inputs. The model predicts a 9.0% decline in electricity use, 2.4 percentage points larger than the reduced-form estimate of 6.6%, while predicting declines of only 0.4% in labor and materials, compared with reduced-form estimates of 1.3%. By imposing a unit elasticity of substitution, $\sigma = 1$, the Cobb–Douglas specification allows plants to substitute away from electricity too easily. It therefore overstates the decline in electricity use while understating the contractions in other inputs and output.

7.3 Amplification Mechanisms

7.3.1 Capital Idleness

We augment our model to address two discrepancies between the reduced-form estimates and the Cobb–Douglas predictions. First, to explain why the Cobb–Douglas model understates

output losses, we allow electricity supply shocks to idle installed capital. Consider a steel plant that operates electric furnaces. When electricity is unavailable, the plant must reduce furnace operating hours, leaving part of its capacity idle. This capital idleness directly reduces output and lowers the marginal products of labor and materials, prompting the plant to reduce both inputs. Capital idleness thus amplifies the output loss beyond the direct effect of reduced electricity use.

We introduce capital idleness for two reasons. First, the literature on energy shocks establishes capital utilization as an important mechanism underlying large fluctuations in aggregate output (see, e.g., Berndt and Wood, 1975; Finn, 2000). Electricity unavailability can similarly amplify output losses by reducing capital utilization. Second, the heterogeneity results in Section 5.4 support this mechanism. Electricity supply shocks cause disproportionately large output losses at more capital-intensive plants and in more capital-intensive industries, whereas their effects do not vary systematically with electricity intensity. This pattern is consistent with capital utilization as the underlying mechanism.

We therefore modify the model so that production depends on effective capital services, $\widetilde{K}$, rather than directly on the installed capital stock, K:

$$Y_{it} = \Omega_i \left[\alpha_K \widetilde{K}_{it}^{\rho} + \alpha_L L_{it}^{\rho} + \alpha_M M_{it}^{\rho} + \alpha_E E_{it}^{\rho}\right]^{1/\rho}. \tag{18}$$

In this augmented model, electricity affects output directly as a production input and indirectly through capital utilization. For example, an electricity constraint can reduce the services provided by a steel plant's electric furnaces without changing its installed furnace capacity.

We formalize this mechanism by defining effective capital services as

$$\widetilde{K}_{it} = \left[(1-\kappa)K_{it}^{\rho} + \kappa\left(v_i E_{it}\right)^{\rho}\right]^{1/\rho}, \qquad v_i \equiv \frac{K_{i,2010}}{E_{i,2010}}, \tag{19}$$

where K_{it} is installed capital, $v_i E_{it}$ is the amount of capital that plant i can operate with its electricity input, and $\kappa \in [0,1]$ governs the dependence of effective capital services on electricity. The normalization $v_i \equiv K_{i,2010}/E_{i,2010}$ implies full capital utilization in 2010 and yields

$$\left.\frac{\partial \ln \widetilde{K}_{it}}{\partial \ln E_{it}}\right|_{t=2010} = \kappa. \tag{20}$$

Using effective capital services ($\widetilde{K}_{it}$) instead of installed capital (K_{it}) changes the baseline input cost shares. The Appendix E-4 describes how we incorporate capital idleness into the

simulation. After this adjustment, the baseline capital and electricity cost shares are

$$\widetilde{s}_{Kj}(\kappa) = (1-\kappa)s_{Kj}, \qquad \widetilde{s}_{Ej}(\kappa) = s_{Ej} + \kappa s_{Kj}. \tag{21}$$

where s_{Kj} and s_{Ej} are the baseline capital and electricity cost shares in industry j, and $\widetilde{s}_{Kj}(\kappa)$ and $\widetilde{s}_{Ej}(\kappa)$ are the corresponding shares after accounting for electricity-dependent capital utilization.

A positive κ therefore shifts κs_{Kj} from the capital cost share to the electricity cost share. Electricity affects output both as a direct production input and by determining how much installed capital can operate. Consequently, higher electricity prices or tighter rationing have stronger effects on unit costs, input demand, and output when capital idleness is more important.

The magnitude of κ is ambiguous a priori. Some machinery may require electricity in nearly fixed proportions, leading electricity shortages to idle substantial capital. However, retrofitting machinery or changing operating practices may limit capital idleness. We therefore estimate κ from the data.

7.3.2 Input Complementarity

Second, to explain why the Cobb–Douglas model overstates the decline in electricity use and understates the declines in labor and materials, we relax its assumption that the elasticity of substitution, σ, equals one. When $\sigma < 1$, electricity is complementary to other inputs, so electricity supply shocks induce plants to reduce labor and materials rather than substitute these inputs for electricity. This limited substitutability produces a smaller decline in electricity use and larger declines in labor, materials, and output, bringing the model predictions closer to the reduced-form estimates.

7.3.3 Calibration

We jointly estimate the elasticity of substitution, σ, and the utilization elasticity, κ, using minimum-distance GMM. The empirical targets are the post-phase-out 2SLS impacts of electricity supply shocks on electricity use, labor, materials, and output. For each candidate pair (σ, κ), we calculate the model-implied responses using the simulation procedure described above. We then construct four moment discrepancies as the differences between the model-implied and reduced-form responses. We select the parameter pair that minimizes the sum of the squared discrepancies. Appendix E-4 presents the detailed procedure.

We obtain $\widehat{\sigma} = 0.57$ and $\widehat{\kappa} = 0.17$, which minimize the discrepancy between the model-implied and reduced-form impacts of electricity supply shocks. At the mean post-phase-out supply shock, the reduced-form estimates imply declines of 6.6% in electricity use, 1.3% in

labor, 1.3% in materials, and 1.7% in output. The calibrated model predicts corresponding declines of 6.7%, 1.4%, 1.4%, and 1.6%. The model therefore closely reproduces the input and output responses.

The estimate $\widehat{\sigma} = 0.57$ indicates limited substitutability between electricity and other inputs. It implies stronger input complementarity than a Cobb–Douglas technology and falls within the range of approximately 0.2–0.8 reported for substitution between energy and other inputs.[19] Comparisons with estimates specific to electricity are more difficult because the empirical evidence remains scarce.[20]

Panel D of Figure 6 varies σ while holding $\kappa = 0.17$ and the baseline cost shares fixed. Lowering σ from its calibrated value of 0.57 to 0.2 strengthens input complementarity. The predicted decline in electricity use falls from 6.7% to 4.4%, while the declines in labor and materials increase from 1.4% to 2.0% and the output loss increases from 1.6% to 2.0%. Conversely, raising σ to 1.5 increases input substitutability. The predicted decline in electricity use rises to 13.8%, while the declines in labor and materials fall to 1.0% and the output loss falls to 1.4%.

The estimate $\widehat{\kappa} = 0.17$ implies that, at the baseline, a 1% reduction in electricity use reduces effective capital services by 0.17%. This utilization channel is quantitatively important because electricity's direct output elasticity is only 0.015, approximately one-tenth of capital's output elasticity of 0.139. Panel E of Figure 6 illustrates the extent to which capital idleness amplifies the effects of electricity supply shocks. Eliminating the utilization channel by setting $\kappa = 0$ reduces the predicted output loss from 1.6% to 0.6%, a reduction of approximately 61%. The corresponding declines in labor and materials fall from 1.4% to 0.5%. Conversely, setting $\kappa = 0.5$ increases the predicted output loss from 1.6% to 3.3%, more than doubling the effect. The corresponding declines in labor and materials increase to 2.9%.

8 Policy Counterfactuals

In this section, we use the structural model to evaluate counterfactual policies that could mitigate the output effects of electricity supply shocks.

[19] Using U.S. manufacturing-plant data and variation in energy prices, Hawkins-Pierot and Wagner (2026) estimates a common elasticity of substitution among capital, labor, and energy of 0.26 across all industries and 0.24 among energy-intensive industries. van der Werf (2008) reports energy substitution elasticities of approximately 0.16–0.62 across industries, while Bretschger and Jo (2024) estimates labor–energy elasticities of 0.61–0.80 using data on French manufacturing firms.

[20] Aggarwal (2026) estimates labor–electricity elasticities of approximately 0.1 in India's formal manufacturing sector and 0.2–0.5 in its informal manufacturing sector.

8.1 Decomposing the Effects of Pricing and Rationing

Electricity supply shocks affected manufacturing plants through higher electricity prices and peak-season rationing. We use the structural model to decompose the effects of these two channels during the post-phase-out period (2013–2018). Their relative importance indicates whether policies should focus on reducing electricity costs or relaxing physical supply constraints.

To separate the price and rationing channels, we compare plant-level predictions for electricity use, labor, materials, and output across three scenarios: the 2010 no-shock baseline, a price-only scenario, and a combined price-and-rationing scenario. The decomposition is

$$\underbrace{\ln Z_i^{\text{sim}}(\overline{p}_r^E, \overline{R}_r) - \ln Z_i^{\text{sim}}(1,0)}_{\text{Total impact}} = \underbrace{\ln Z_i^{\text{sim}}(\overline{p}_r^E, \overline{R}_r) - \ln Z_i^{\text{sim}}(\overline{p}_r^E, 0)}_{\text{Rationing contribution}} + \underbrace{\ln Z_i^{\text{sim}}(\overline{p}_r^E, 0) - \ln Z_i^{\text{sim}}(1,0)}_{\text{Price contribution}}. \tag{22}$$

where $Z_i^{\text{sim}}(\cdot,\cdot)$ denotes plant i's simulated annual outcome, with the first argument representing the relative electricity price and the second representing peak-season rationing intensity. We normalize the 2010 baseline to $(1,0)$. For region r, $\overline{p}_r^E$ is the predicted post-phase-out relative price, and $\overline{R}_r$ is the mean electricity-saving target used as the rationing intensity.

The first term on the right-hand side measures the rationing contribution by comparing the combined and price-only scenarios. The second term measures the price contribution by comparing the price-only scenario with the no-shock baseline. By construction, the two contributions sum to the total impact. This ordering assigns their interaction to the rationing contribution.

Table 7 shows that pricing accounts for most of the model-implied decline. During the post-phase-out period, the structural model predicts a 6.72% decline in electricity use. The price contribution is 6.23 percentage points, or 92.7% of the total decline, while the rationing contribution is 0.49 percentage points, or 7.3%. The model also predicts declines of 1.39% in labor, 1.39% in materials, and 1.57% in output. Pricing accounts for 1.30, 1.30, and 1.46 percentage points of these declines, respectively, corresponding to 93.0% of each total effect. Rationing accounts for the remaining 7.0%. Although the physical-rationing constraint binds for 85.7% of plants during summer, it applies for only 90 days per year and is generally small in magnitude. Consequently, rationing accounts for only about 7% of the annualized effects on electricity use, labor, materials, and output.

These results indicate that the nuclear shutdown primarily affected manufacturing plants through higher electricity prices. Utilities replaced nuclear generation with more expen-

sive fossil generation, raising generation costs. Binding quantity constraints played a much smaller role. This pattern differs from the residential sector, where non-price interventions can meaningfully reduce electricity demand during scarcity (e.g., Reiss and White, 2008; He and Tanaka, 2023).

8.1.1 Elasticity Benchmarks

We report elasticities with respect to electricity-price changes using both the reduced-form estimates and the structural model. These elasticities provide benchmarks for the counterfactual analysis.

The reduced-form analysis implies an own-price elasticity of -0.80 for electricity, cross-price elasticities of -0.15 for both labor and materials, and an output elasticity of -0.20 (Table E4, Column (1)). We calculate each elasticity as the ratio of the main reduced-form effect of the supply shock on the outcome to its estimated effect on the log electricity price. At the mean post-phase-out supply shock, the 2010-total-salary-weighted mean predicted log-price change is 8.4 log points, corresponding to an average electricity-price increase of approximately 9.0% in levels. This estimated price response captures only the change in the market price and does not include the shadow price associated with physical rationing.

The structural model yields corresponding price-only elasticities of -0.74, -0.15, -0.15, and -0.17 (Column (2)). We calculate the plant-level responses using the model's closed-form expressions. For example, the own-price elasticity of electricity demand is

$$\epsilon_{E,i} \equiv \frac{\partial \ln E_i}{\partial \ln p^E} = -\sigma + (\sigma - \eta) s^C_{E,ij},$$

where $s^C_{E,ij}$ is electricity's share of the plant's unit cost. We then aggregate the plant-level elasticities using 2010 total-salary weights. Appendix E-3.2 describes the derivation.[21]

8.2 Counterfactual: Regional Market Integration

We use the estimated production framework to evaluate regional market integration as a supply-side intervention that can mitigate the economic costs of electricity supply disruptions. Section 6.1 shows that manufacturing plants have limited capacity to adapt to supply shocks. We therefore focus on a policy that improves electricity-market functioning and lowers electricity prices.

[21] Our own-price elasticity estimates are consistent with the quasi-experimental evidence and are closest to von Graevenitz and Rottner (2024), who report estimates of approximately -0.4 to -0.6. Previous studies report industrial own-price elasticities ranging from -0.09 to -0.20 (Gerster and Lamp, 2024) and values as large as -1.5 in settings with substantial price shocks (Martin et al., 2014).

Japan's electricity market has historically been segmented because regulations limited interregional electricity transmission. This segmentation can create dispatch inefficiencies: high-marginal-cost generators may serve local demand while lower-cost capacity elsewhere remains idle (Hausman, 2025). It also limits reserve sharing, requiring regions to maintain redundant backup capacity and raising system-wide costs. The potential gains from greater market integration have therefore attracted substantial policy attention. We quantify the benefits of relaxing these constraints through interregional trade and more efficient reserve sharing.

We quantify the resulting changes in output and producer surplus attributable to electricity supply shocks. In particular, we approximate the policy-induced change in aggregate producer surplus as

$$\Delta\Pi \approx \sum_i \Pi_i\left(p^E_{\text{base}}\right)\ (1-\eta)\, s^C_{E,i}\, \Delta \ln p^E, \tag{23}$$

where $s^C_{E,i}$ is plant i's electricity cost share and $\Delta \ln p^E$ is the policy-induced change in the electricity price. This approximation follows from the model's electricity-price elasticities; Appendix E-3 describes the counterfactual simulation procedure.

Estimating these effects requires counterfactual electricity prices under market integration (Appendix F-0.1). We estimate regional supply curves and solve for integrated-grid equilibrium prices, allowing lower-cost generators to serve a wider market and enabling reserve sharing. Full integration reduces the national average log electricity price by 2.8 log points relative to the observed scenario.[22] The mean log-price increase relative to the no-shock baseline falls from 8.4 to 5.7 log points, corresponding to level increases of approximately 9.0% and 5.8%, respectively. Price changes remain heterogeneous, with large declines in high-cost demand centers such as Tokyo and offsetting increases in low-cost exporting regions (Table F1, Column (2)).

Table 7 summarizes the simulation results. Under the price-only structural baseline (the price-contribution row of Panel B), electricity use declines by 6.2%, while labor and materials each decline by 1.3%. Real output and producer surplus fall by 1.5% and 1.2%, respectively. Evaluated at 2010 levels, the output decline corresponds to an annual loss of 4,231 billion Japanese yen, compared with the reduced-form estimate of 4,920 billion Japanese yen.[23] Producer surplus falls by 677 billion Japanese yen.

[22]The implied price effects are broadly consistent with existing evidence on market integration. Gonzales et al. (2023) estimate that Chile's 2017 integration reform reduced generation costs by 3–7% in the short run, and by 8–18% when accounting for investment responses. Wolak (2015) finds that relieving transmission congestion in New Zealand lowers wholesale prices by roughly 3% on average. Our estimates are smaller than those in Ryan (2021), who find larger gains in settings with severe transmission shortages.

[23]These correspond to approximately 45–55 billion US dollars, using an exchange-rate range of 80–95 Japanese yen per US dollar (around 2010).

Market integration with reserve sharing substantially mitigates these costs (Panel C). It reduces the decline in electricity use from 6.2% to 4.2% and the output decline from 1.5% to 1.0%. Annual output losses fall from 4,231 billion Japanese yen to 2,864 billion Japanese yen, while producer-surplus losses fall from 677 billion Japanese yen to 458 billion Japanese yen. Both losses decline by approximately 32.3% relative to the baseline.

These results indicate that limited interregional coordination amplified the costs of the post-Fukushima electricity supply shock. Greater market integration recovers nearly one-third of the price-induced losses by improving system-wide dispatch and reserve sharing.

9 Conclusion

This paper shows that the persistent electricity supply shock following the Fukushima accident imposed sizable costs on Japanese manufacturing. A 6% decline in electricity use reduced revenue by 1.7%, implying annual losses of roughly 4.9 trillion Japanese yen despite electricity's 2% expenditure share. Plants showed little adjustment through technology, product mix, fuel substitution, or production reallocation; self-generation protected only a small subset of producers.

Our structural model explains these losses through strong input complementarity and capital idleness and closely matches the reduced-form estimates. The counterfactuals show that higher electricity prices drove most of the output decline and that greater grid integration could recover nearly one-third of the baseline losses. The results highlight the importance of affordable, efficiently allocated electricity for industrial resilience.

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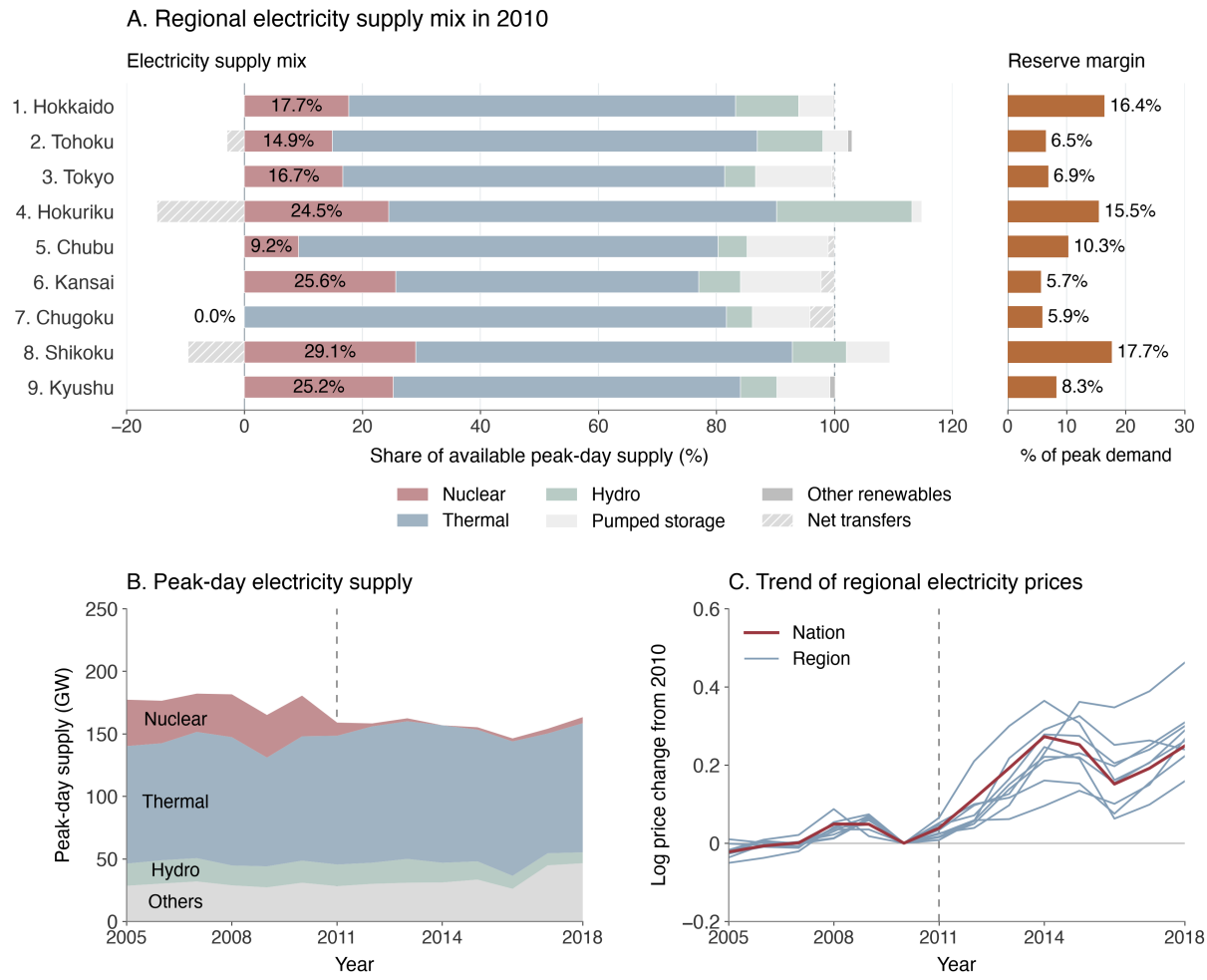
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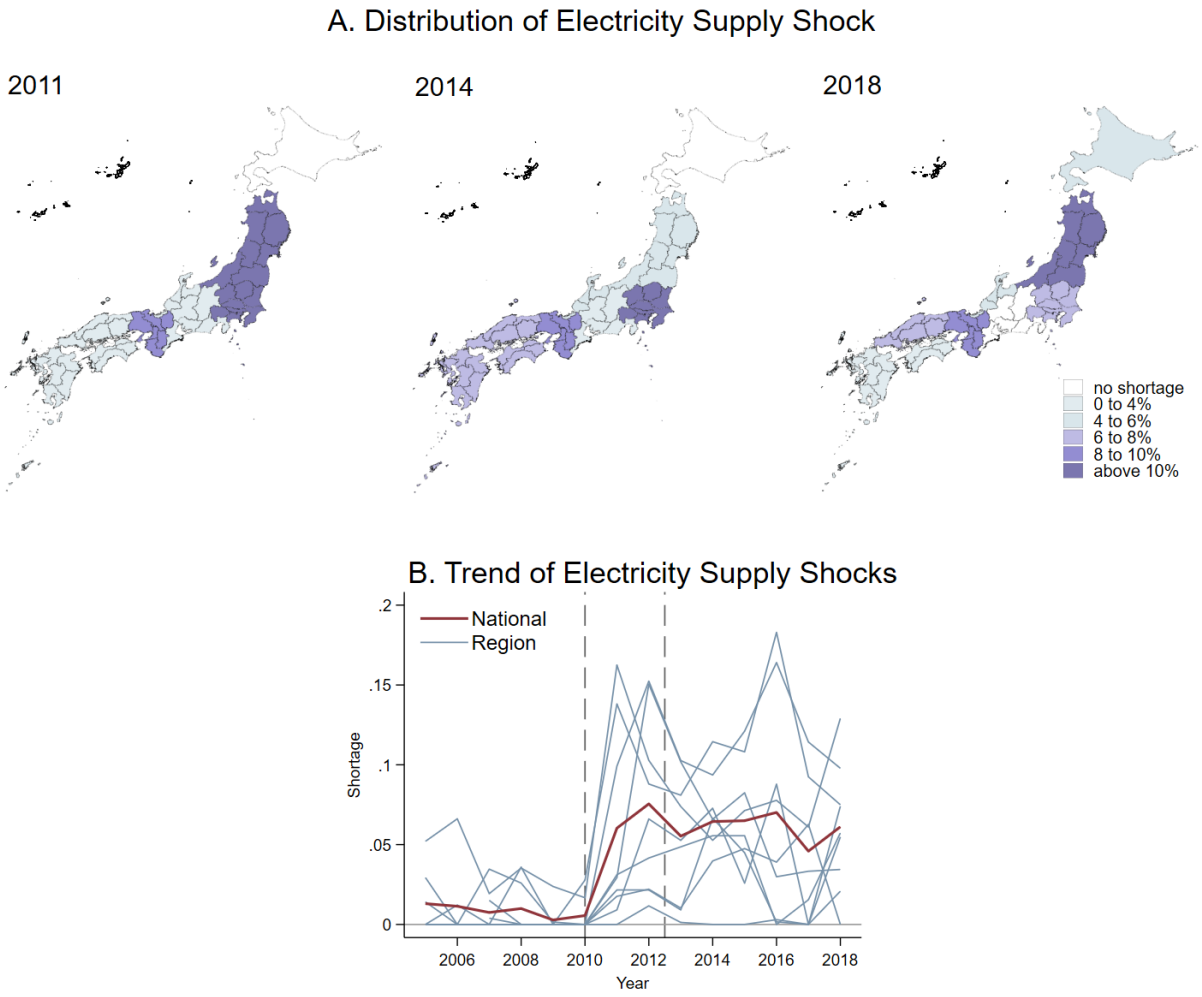
Figures and Tables

Figure 1: Nuclear shutdown and electricity supply shocks



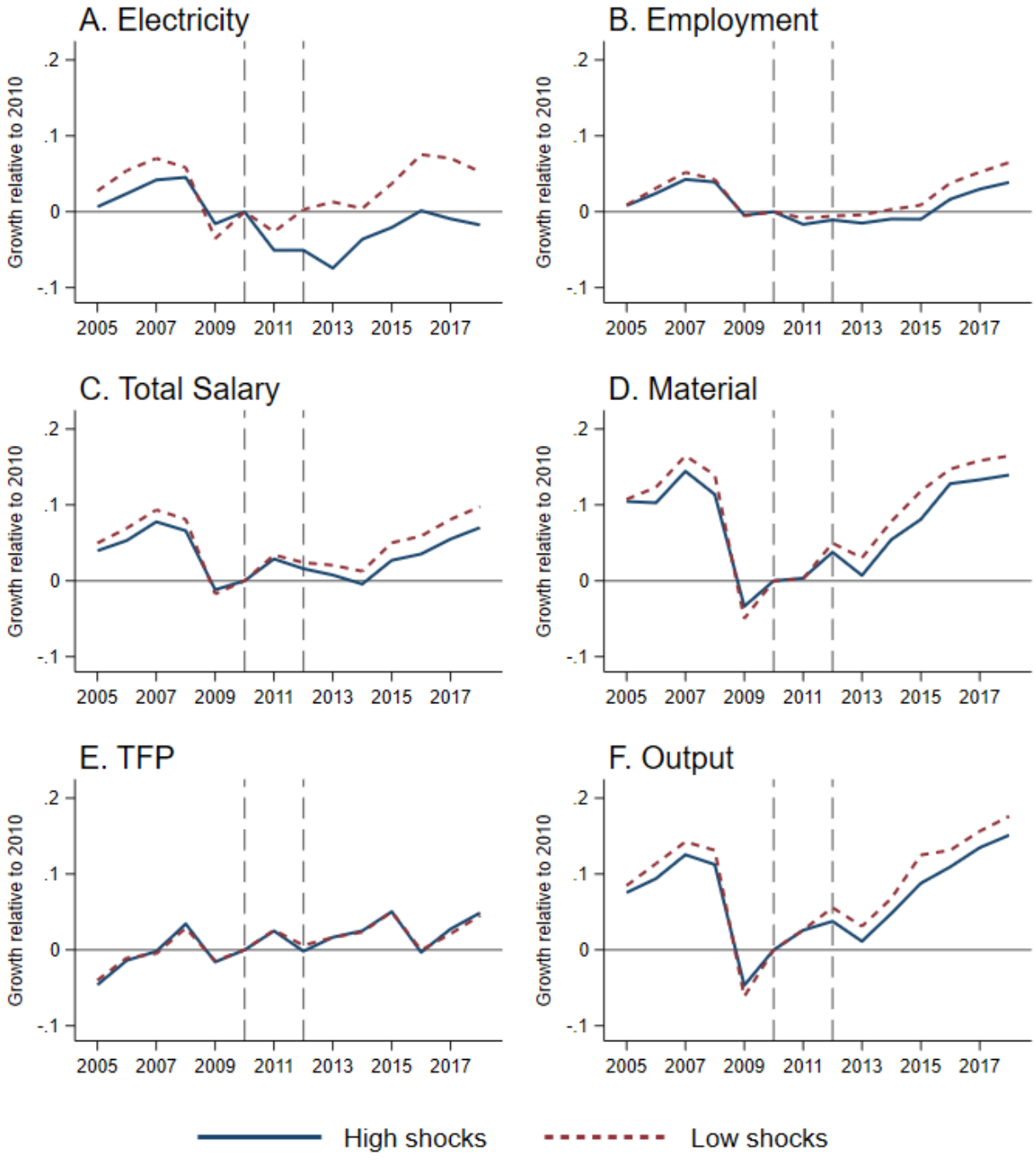
Notes: Panel A shows each source's share of available peak-day electricity supply and reserve margins in 2010 for the nine mainland regions. Reserve margins equal available supply minus peak demand, divided by peak demand. Hatched bars show net transfers; negative values indicate net exports. Pumped storage is separate from conventional hydropower. Regional peak dates may differ, and source totals may differ slightly because of rounding. Sources: METI electricity supply–demand data (ESDV, 2012) and the paper's peak-demand series. Panel B shows peak-day electricity supply by source in GW. Values for 2005–2015 sum the nine regions' individual peaks; values for 2016–2018 reconstruct source components by scaling supply-capacity shares to national coincident peak demand. Others includes purchases, other sources, and net adjustments. Panel C shows log electricity-price changes relative to 2010. The bold line represents the national trend, and thin lines represent individual regions. In Panels B and C, the vertical dashed line marks 2011.

Figure 2: Distribution and trend of electricity supply shocks



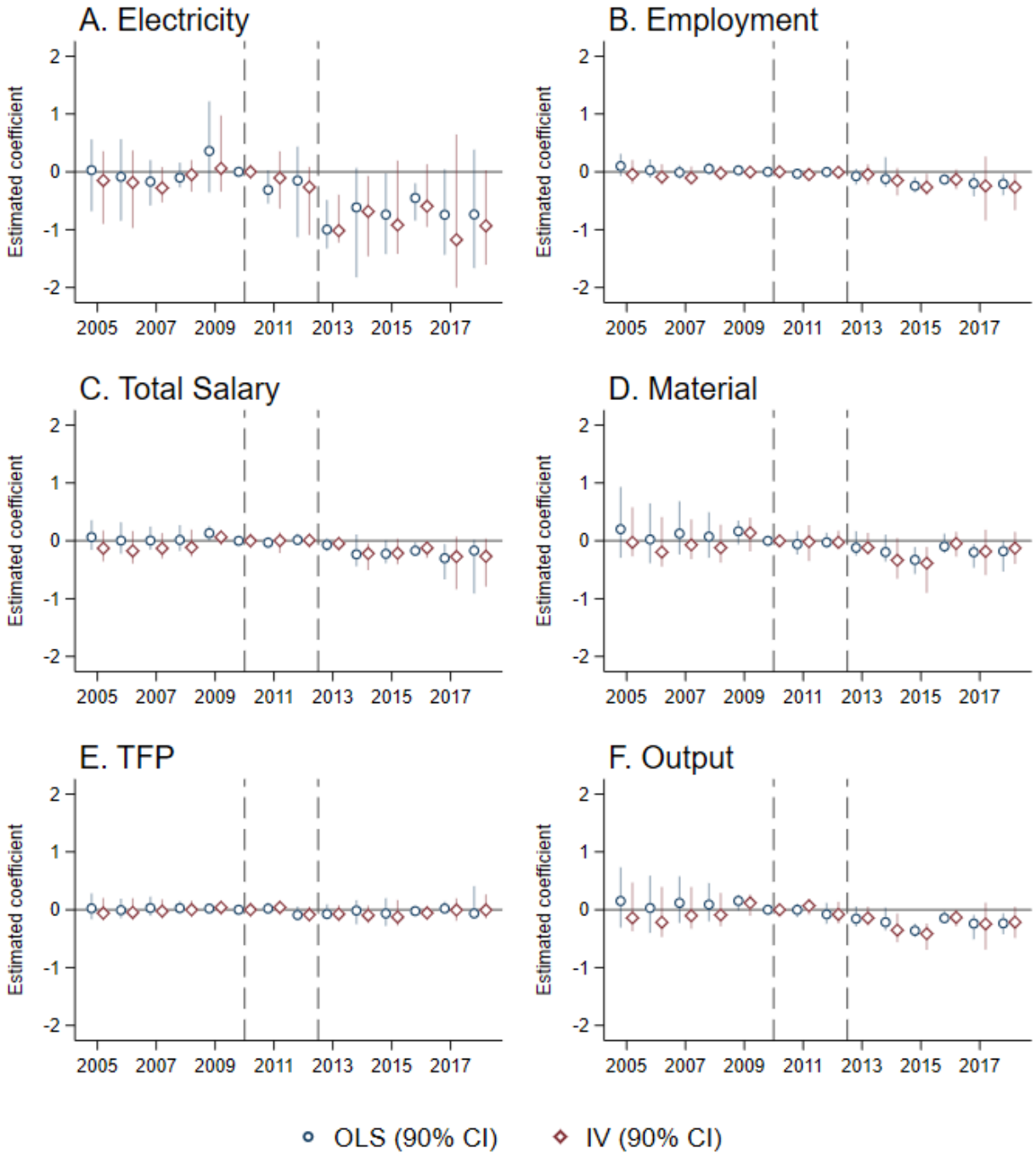
Notes: Panel A presents the geographic distribution of our measure of electricity supply shocks from 2011 to 2018. Panel B shows the corresponding trends, where the bold line represents the national average and the thin gray lines represent individual regions. See the main text for the definition of electricity supply shocks.

Figure 3: Trends of various measures of manufacturing performances between high-shock and low-shock regions



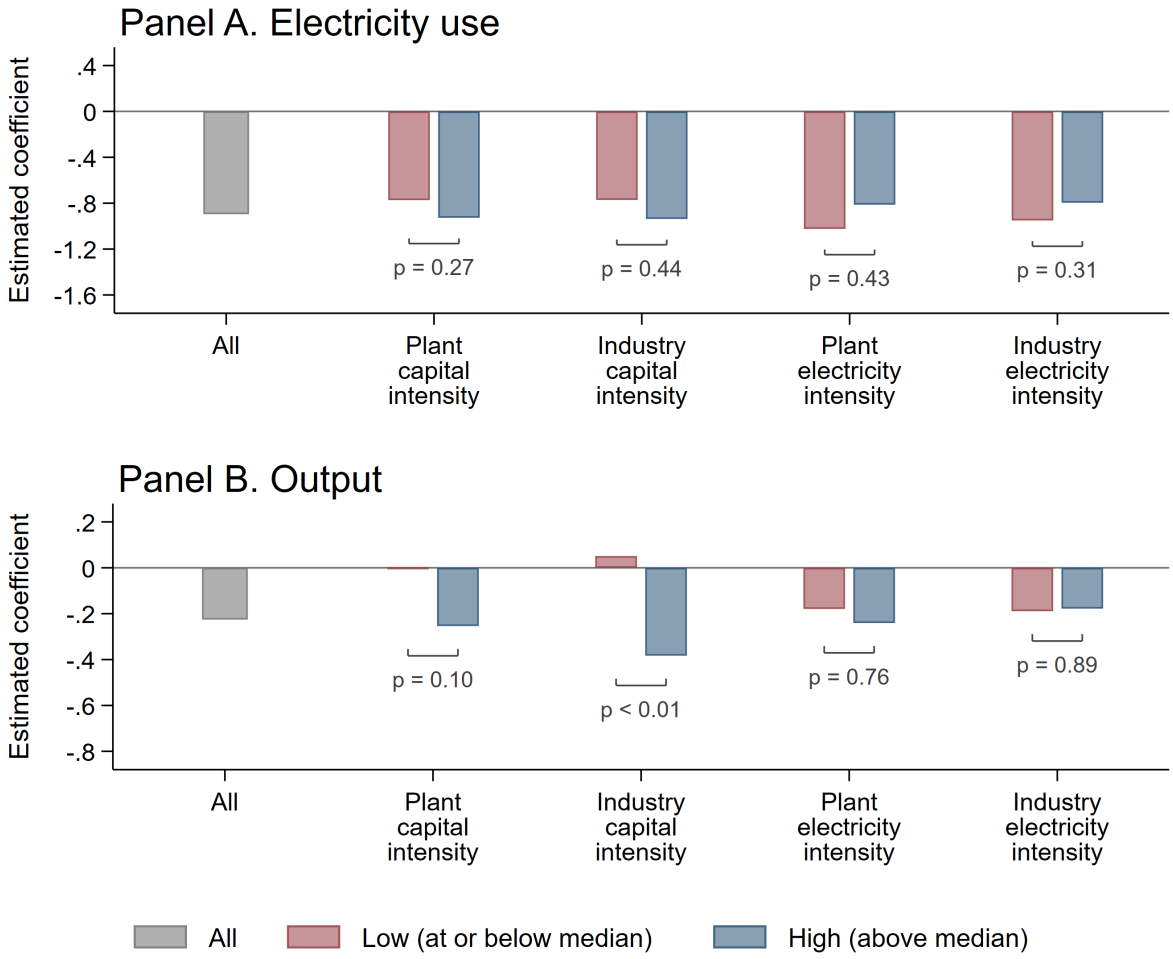
Notes: We classify regions with mean supply shocks during 2013–2018 at or above the median across the nine regions as high-shock regions and the remaining regions as low-shock regions. Outcomes are normalized by their 2010 values, so the y-axis represents changes relative to 2010.

Figure 4: Event study results of effects of supply shock on various manufacture's activities



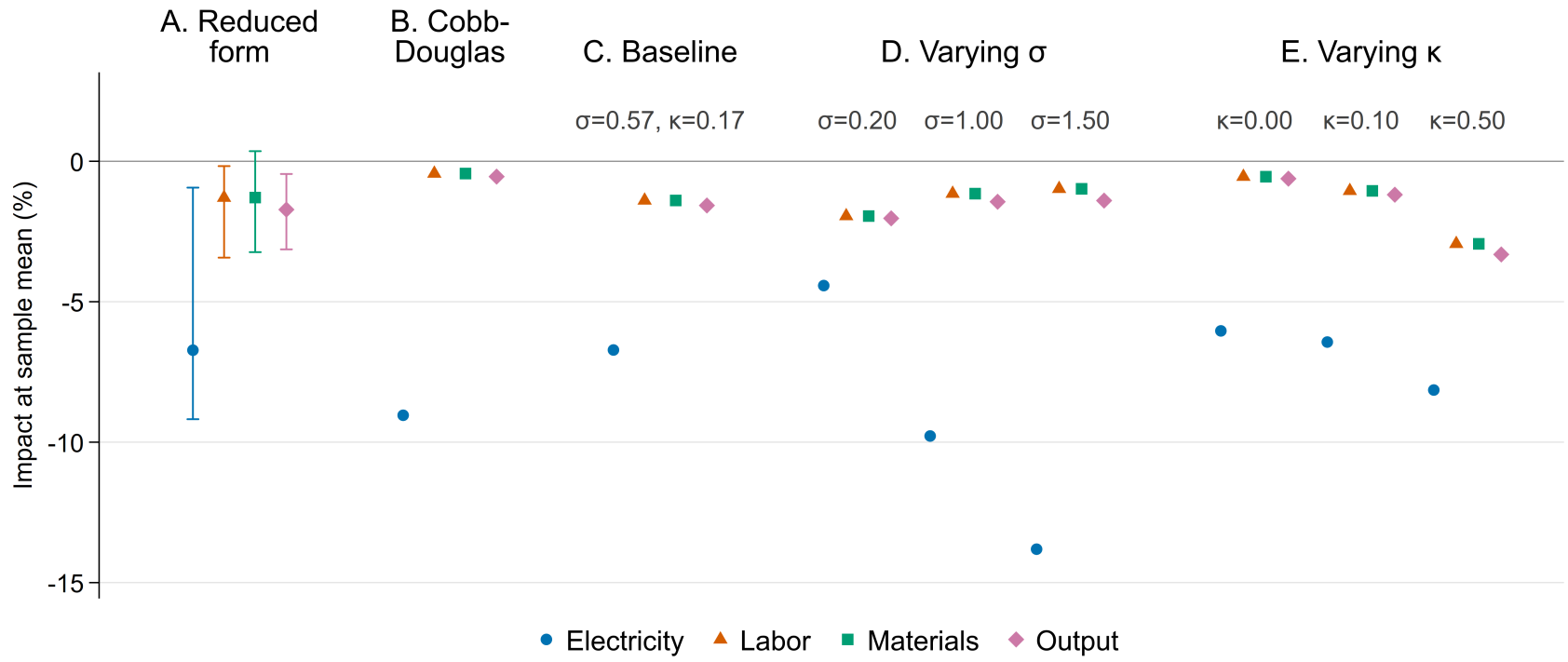
Notes: The figure reports event-study estimates from equation (6). Blue circles denote OLS estimates, and red diamonds denote 2SLS estimates. Vertical bars show 90% confidence intervals obtained using a wild cluster bootstrap with clustering by region. Confidence intervals extending below -2 are truncated at -2. For pre-accident years, we use each region's mean supply shock during the post-phase-out period. All regressions include three-digit-industry-by-year fixed effects and use 2010 log-salary weights.

Figure 5: Heterogeneous impacts of electricity supply shocks



Notes: The figure reports 2SLS coefficients from equation (5) for electricity use (Panel A) and output (Panel B). Gray bars show the full-sample estimates. Red and blue bars show estimates for plants at or below and above the median, respectively, of the indicated plant- or three-digit-industry characteristic measured in 2010. Brackets report wild-cluster-bootstrap p-values for tests of equality between the two subgroup coefficients, clustering by region. All regressions include year-by-three-digit-industry fixed effects and use 2010 log-salary weights.

Figure 6: Reduced-Form Estimates and Structural Simulations



Notes: The figure compares reduced-form impacts at the weighted mean post-phase-out supply shock with structural predictions. Reduced-form whiskers show 90% confidence intervals. Panel B reports the Cobb–Douglas benchmark, and Panel C reports the calibrated baseline with $\sigma = 0.57$ and $\kappa = 0.17$. Panel D varies $\sigma \in \{0.20, 1.00, 1.50\}$ while holding $\kappa = 0.17$ and baseline cost shares fixed. Panel E varies $\kappa \in \{0.00, 0.10, 0.50\}$ while holding $\sigma = 0.57$. Within every specification, markers appear in the order electricity, labor, materials, and output. Generator owners retain their simulated grid-electricity response, while their structural non-electricity and output responses are set to zero.

Table 1: Summary Statistics

	Mean (1)	S.D. (2)
Panel A. Plant Characteristics		
Revenue (million Japanese yen)	4,999	11,313
Employment (headcount)	127	166
Capital stock (million Japanese yen)	1,035	2,485
Investment (million Japanese yen)	185	532
Export share (%)	2.8	11
Multi-plants (yes = 1)	0.42	0.49
Panel B. Revenue shares		
Labor	0.22	0.14
Capital	0.033	0.042
Materials	0.51	0.20
Electricity	0.020	0.023
Panel C. Plant Energy Usage		
Fuel input for production (TJ)	30	114
Fuel input for self-generation (TJ)	4.2	31
Self-generated electricity (TJ)	3.6	24
Grid electricity purchased (TJ)	20	55
Self-generation ownership (yes = 1)	0.13	0.34

Notes: Panels A and B use 437,456 plant-year observations from the Census of Manufactures; the export-share row uses 435,855 observations because of missing export data. Expenditure shares in Panel B are calculated as the ratio of input expenditure to plant revenue. Panel C uses 380,682 plant-year observations from the Energy Consumption Statistics and the Petroleum and Other Fuels Monthly Statistics. All panels exclude Okinawa. Monetary values are deflated to 2010 prices.

Table 2: Estimated Output Elasticities and Productivity Dispersion

	Structural Estimates			Benchmark
	Mean (1)	p25 (2)	p75 (3)	OLS (4)
Panel A. Output elasticities				
Materials (θ_M)	0.446	0.386	0.506	0.492
Electricity (θ_E)	0.015	0.008	0.017	0.068
Labor (θ_L)	0.354	0.259	0.432	0.423
Capital (θ_K)	0.139	0.083	0.170	0.042
Panel B. Productivity dispersion				
75/25 productivity ratio	1.512	1.373	1.556	1.375
90/10 productivity ratio	2.350	1.940	2.424	1.953

Notes: Panel A summarizes the distribution of output elasticities estimated using the Gandhi, Navarro, and Rivers (2020) method. Columns (1)–(3) report the mean and the 25th and 75th percentiles of the structural estimates across 132 industries with converged, strictly positive elasticities. Column (4) provides the mean benchmark across the same industries based on industry-specific OLS regressions of log revenue on log inputs. Materials and electricity are treated as flexible inputs and estimated in the first step from expenditure-share equations. Labor and capital are treated as quasi-fixed inputs and estimated jointly in the second step by GMM. Panel B reports productivity-percentile ratios calculated within each industry using plant-year observations from 2005–2010 and then summarized across industries. Structural productivity is measured in levels as $\exp(\widehat{\omega}_{it} + \widehat{\epsilon}_{it})$; the OLS benchmark is the exponentiated uncorrected OLS residual. These productivity measures are implied by the output-elasticity estimates and do not impose a particular CES elasticity of substitution.

Table 3: First-Stage Regression: Impact of Supply Composition on Supply Shocks

	Dep Var: Supply Shocks	
	Phase-out	Post-phase-out
	(1)	(2)
Predicted nuclear supply growth	-0.45	-0.26
	(0.00)***	(0.00)***
Reserve margin in 2010	-1.05	-0.82
	(0.00)***	(0.00)***
Year FE	Yes	Yes
Observations	18	54

Notes: The table reports first-stage regression estimates. The dependent variable is the electricity supply shock. Predicted nuclear supply growth is defined as $Sh^{Nuc}_{r,2010} \times g^{Nuc}_{t,-r}$, where $Sh^{Nuc}_{r,2010}$ is region r's nuclear generation share in 2010 and $g^{Nuc}_{t,-r}$ is the leave-one-out national growth rate in nuclear generation. Column (1) covers the nuclear phase-out period (2011–2012), and Column (2) covers the post-phase-out period (2013–2018). P-values are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 4: Effects of Supply Shocks on Inputs and Outputs

	Dep Var: $\Delta \ln(Y) = \ln(Y) - \ln(Y_{2010})$					
	Electricity (1)	Labor (2)	Total Salary (3)	Materials (4)	TFP (5)	Output (6)
Panel A. The effects of supply shocks using OLS						
Supply Shocks	-0.24	-0.02	-0.01	-0.04	-0.03	-0.04
(Phase-out)	[0.11]	[0.42]	[0.72]	[0.27]	[0.73]	[0.23]
Supply Shocks	-0.63**	-0.15**	-0.18**	-0.15	-0.03	-0.19***
(Post-phase-out)	[0.03]	[0.02]	[0.04]	[0.13]	[0.55]	[0.00]
Panel B. The effects of supply shocks using 2SLS						
Supply Shocks	-0.21	-0.03	0.01	-0.01	-0.04	-0.02
(Phase-out)	[0.36]	[0.11]	[0.91]	[0.82]	[0.69]	[0.61]
Supply Shocks	-0.89*	-0.17	-0.17*	-0.17	-0.07	-0.23**
(Post-phase-out)	[0.06]	[0.12]	[0.06]	[0.18]	[0.44]	[0.04]
Year-by-Industry FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table reports regression estimates. The dependent variables are log differences from 2010. TFP is the plant-level production residual, $\widehat{\omega}_{it} + \widehat{\epsilon}_{it}$, constructed using industry-specific input elasticities estimated from the 2005–2010 data. The phase-out row uses data from 2005–2010 and 2011–2012. The post-phase-out row uses data from 2005–2010 and 2013–2018. All regressions include year-by-industry fixed effects, where industries are defined at the three-digit level, and use 2010 log labor weights. P-values from wild cluster bootstrap using regions as clusters are reported in brackets. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 5: Effects of Supply Shocks on Plant Behavior

	Product-level		Outsourcing	
	Switching (1)	Intensity (2)	Outsourcing (3)	Share in Mat. (4)
Panel A. Effects of supply shocks using OLS				
Supply shocks (Phase-out)	0.09 [0.23]	0.02 [0.50]	-0.02 [0.88]	0.00 [0.96]
Supply shocks (Post-phase-out)	0.03 [0.72]	0.03 [0.27]	-0.06 [0.86]	-0.01 [0.76]
Panel B. Effects of supply shocks using 2SLS				
Supply shocks (Phase-out)	0.07 [0.33]	0.02 [0.32]	-0.02 [0.95]	-0.05 [0.51]
Supply shocks (Post-phase-out)	0.07 [0.52]	0.03 [0.44]	-0.36 [0.37]	-0.08 [0.46]
Year-by-industry FE	Yes	Yes	Yes	Yes

Notes: The table reports regression estimates. Column (1) uses a dummy variable for product switching. Column (2) measures product-level electricity intensity. Column (3) uses log differences in outsourcing relative to 2010. Column (4) uses log differences in the outsourcing share in materials relative to 2010. The phase-out row uses data from 2005–2010 and 2011–2012. The post-phase-out row uses data from 2005–2010 and 2013–2018. All regressions include year-by-industry fixed effects, where industries are defined at the three-digit level, and use 2010 log-labor weights. P-values from wild cluster bootstrap using regions as clusters are reported in brackets. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 6: Effects of Supply Shocks on Energy Usage

	Dep Var: $\Delta \ln(Y) = \ln(Y) - \ln(Y_{2010})$			
	Fuel Consumption for:			
	Production (1)	Self-generation (2)	Electricity Generated (3)	Generator ownership (4)
Panel A. Effects of supply shocks using OLS				
Supply shocks (Phase-out)	-0.12 [0.39]	0.98 [0.28]	0.34 [0.40]	0.23*** [0.00]
Supply shocks (Post-phase-out)	-0.26 [0.19]	3.29** [0.04]	1.52* [0.07]	0.22** [0.04]
Panel B. Effects of supply shocks using 2SLS				
Supply shocks (Phase-out)	-0.12 [0.35]	1.25 [0.23]	0.51 [0.46]	0.23*** [0.01]
Supply shocks (Post-phase-out)	-0.35 [0.53]	4.23** [0.03]	1.84* [0.10]	0.30*** [0.00]
Year-by-industry FE	Yes	Yes	Yes	Yes

Notes: This table uses the ECS data and reports regression estimates. Columns (1) and (2) report fuel consumption for production and self-generation, respectively. Column (3) reports self-generated electricity. Column (4) reports generator ownership, defined by any positive self-generation. Ownership begins in the first year a plant reports positive self-generation and persists thereafter. The phase-out row uses data from fiscal years 2007–2010 and 2011–2012. The post-phase-out row uses data from fiscal years 2007–2010 and 2013–2018. All regressions include year-by-industry fixed effects, where industries are defined at the two-digit level, and use 2010 employment weights. P-values from wild cluster bootstrap using regions as clusters are reported in brackets. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 7: Structural Simulation Impacts and Counterfactuals

	Post-phase-out impacts				Monetary losses (billion Japanese yen)	
	Electricity (1)	Labor (2)	Materials (3)	Output (4)	Output (5)	Profit (6)
Panel A. Reduced-form impacts at the mean shock						
Reduced-form estimate	-6.6%	-1.3%	-1.3%	-1.7%	-4,920	–
Panel B. Structural estimation						
Total impact	-6.7%	-1.4%	-1.4%	-1.6%	-4,546	-727
Price contribution	-6.2%	-1.3%	-1.3%	-1.5%	-4,231	-677
(Share of total impact)	(92.7%)	(93.0%)	(93.0%)	(93.0%)	(93.0%)	(93.0%)
Rationing contribution	-0.5%	-0.1%	-0.1%	-0.1%	-316	-50
(Share of total impact)	(7.3%)	(7.0%)	(7.0%)	(7.0%)	(7.0%)	(7.0%)
Panel C. Counterfactual analysis						
Market integration	-4.2%	-0.9%	-0.9%	-1.0%	-2,864	-458

Notes: Columns (1)–(4) report annualized log impacts during the post-phase-out period, expressed in percent. Columns (5) and (6) report annual aggregate output and profit losses in billion constant 2010 Japanese yen. Panel A reports the main reduced-form IV estimates multiplied by the mean shock. Panel B reports predictions from the preferred dynamic-labor structural model. The total impact compares predicted electricity prices and seasonal rationing with the 2010 baseline. The price contribution changes predicted prices while holding rationing at the no-rationing baseline. The rationing contribution is the difference between the total and price-only impacts and therefore includes their interaction. Panel C compares the price-only baseline reported in the price-contribution row of Panel B with full market integration and reserve sharing. Percentage responses are weighted by 2010 total salary. Aggregate losses apply these weighted responses to aggregate 2010 output. Contribution shares are calculated from underlying unrounded estimates. Generator owners retain their simulated grid-electricity response, while their labor, materials, output, and profit responses are set to zero based on the heterogeneous IV evidence.

ONLINE APPENDIX

Electricity Supply Shocks and Manufacturing: Evidence from the Fukushima Nuclear Accident

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A Data Appendix

A-1 Census of Manufactures

A-1.1 Sampling

We use establishment-level data from Japan's Census of Manufactures (CMF) for 2005–2018. The CMF surveys manufacturing establishments with at least four employees. It administers a simplified questionnaire to establishments with 4–29 employees and a detailed questionnaire to those with 30 or more employees. We use the detailed questionnaire because it reports the output, input, inventory, and fixed-asset information required for our analysis.

In 2011 and 2015, the Economic Census for Business Activity collected the corresponding manufacturing data in place of the CMF (METI, 2012; Ministry of Internal Affairs and Communications and Ministry of Economy, Trade and Industry, 2012, 2016). The Ministry of Internal Affairs and Communications and METI jointly conducted the Economic Census. We use the official longitudinal identifier to link the 2011 records to the CMF. For 2015, we link records across years using normalized plant names within prefecture–municipality cells. Analyses that require within-plant changes or a 2010 baseline retain only establishments with valid longitudinal links.

We assign each establishment to the service territory of one of Japan's ten regional electricity utilities. The prefecture identifies the utility for most establishments. For establishments near utility boundaries that cross prefectural borders, we use municipality and postal codes. We exclude Okinawa because its isolated electricity system does not provide variation comparable to that of the nine interconnected utility regions.

A-1.2 Construction of Electricity Use

The CMF reports expenditure on electricity purchased from the grid but does not report physical electricity purchases, plant-specific electricity prices, or electricity generated on site. We use the ECS described in Appendix A-2 to measure on-site generation.

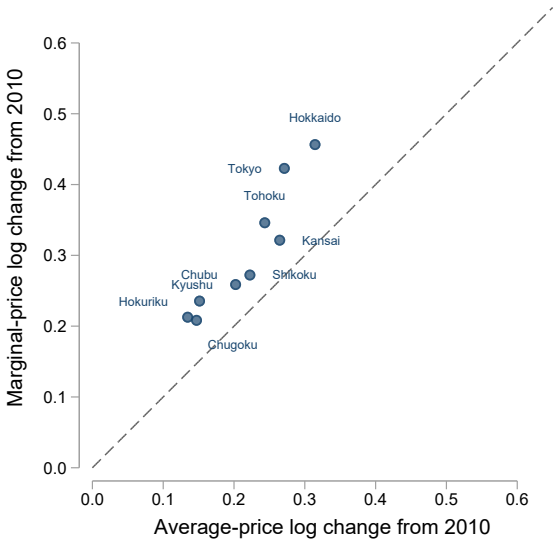
We proxy for grid electricity purchases by dividing each plant's electricity expenditure by the average-price index for its utility region and year, normalized to one in 2010. We calculate the regional average price as the utility's electricity revenue divided by the quantity of electricity it sells, using data from FEPC (FEPC, 2026). The resulting variable measures grid-electricity expenditure in 2010-price units, which we use as a proxy for grid electricity purchases; it does not measure electricity in physical units such as kilowatt-hours. This procedure requires the regional index to capture the common proportional component of plant-level price changes, but it does not require plants within a region to pay identical price levels.

We use the average-price index both to deflate electricity expenditure and to construct the electricity-price changes used in the simulations because the CMF does not report plant-specific tariff schedules or contracted demand. To assess whether the average-price index captures price changes similar to those measured by a marginal price, we construct a marginal-price measure for a representative high-voltage industrial contract. Japanese electricity bills generally consist of four components: a monthly demand charge, a volumetric energy charge, a monthly fuel-cost adjustment, and a renewable-energy surcharge (Agency for Natural Resources and Energy, 2024). These charges follow regional or national schedules and may vary by month.[1] We construct the marginal-price measure using the volumetric charge, fuel-cost adjustment, and renewable-energy surcharge. We exclude the demand charge because it does not vary with electricity consumption when contracted demand remains fixed.

Figure A1 compares the average- and marginal-price measures across the nine utility regions. For each region-year and price measure, we calculate the log difference from the region's 2010 level and then average these differences over 2013–2018. The correlation across the nine regions is 0.913. This descriptive comparison does not establish whether plants respond to average or marginal prices. It shows that the two measures capture similar cross-regional price changes.

[1] The renewable-energy surcharge is common nationwide within each year.

Figure A1: Average and Marginal Electricity-Price Changes



Notes: Each point represents one of the nine utility regions ($N = 9$). Both axes report the region's mean log price difference from its 2010 level during 2013–2018. The horizontal axis uses the average-price index described in the text, and the vertical axis uses the constructed tax-exclusive marginal price. The dashed line is the 45-degree line. The correlation across regions, weighted by plants' total salaries in 2010, is 0.913.

A-1.3 Deflators and Capital Rental Costs

We construct the remaining production variables as follows. We measure nominal output as total receipts adjusted for changes in inventories of finished and semi-finished goods. Nominal materials consist of total intermediate-input expenditure excluding purchased electricity, adjusted for changes in inventories of raw materials and fuels. Our primary labor measure is total cash wages and salaries; we separately examine the number of persons engaged as a physical measure of employment. We measure installed capital using the reported beginning-of-year book value of tangible fixed assets.

We convert monetary variables to a consistent tax-inclusive basis before deflation. We deflate output and materials using industry-specific output and input price indexes from the Bank of Japan (Bank of Japan, 2026). We deflate total cash wages and salaries using the Consumer Price Index from the Statistics Bureau of Japan (Statistics Bureau of Japan, 2026). We deflate capital stock using capital-stock deflators constructed from national capital-stock series published by the Cabinet Office (Cabinet Office, 2026).

Finally, we calculate capital rental costs by multiplying each plant's deflated beginning-of-year capital stock by an industry-year rental rate. We measure the rental rate as nominal

capital cost divided by real net capital stock in the JIP database (Cabinet Office, 2023). Because the CMF, price-index, and JIP data use different industry classifications, we harmonize them primarily at the two-digit industry level.

A-1.4 Multi-plant Firms and Product Switching

We use firm identifiers in the CMF to examine whether firms reallocate activity across establishments in response to localized electricity supply shocks (Section 6.1). For each firm-region-year, we aggregate electricity use and output across the firm's establishments outside the focal region. We then calculate the log change in each outside-region total relative to the corresponding 2010 total. For example, when Tokyo is the focal region, we compare a firm's electricity use and output at all of its non-Tokyo establishments with their 2010 levels. These measures are defined for firms with a valid firm identifier and positive current and 2010 outside-region totals.

We also use product-level CMF records to measure product switching and changes in product mix (Section 6.1). The product-switching sample consists of plants observed in 2010 and in the comparison year. We define the switching indicator as one if the plant's set of reported products differs from its 2010 set because it added or dropped at least one product. To construct product-level electricity intensity, we first calculate the electricity-use-to-sales ratio for each single-product plant in 2010 and take the median ratio within each product. We then assign these product-specific intensities to each plant's reported products and calculate their sales-weighted average in each year. This product-mix intensity measure is available for plant-years in which every reported product has an estimated intensity.

A-1.5 Entry and Exit

We use publicly available aggregate plant-count data from METI's 2006, 2009, 2014, and 2016 Census and Census of Manufactures (CMF) publications to examine plant turnover. We rely on these aggregate data because our restricted-access microdata cover only plants with 30 or more employees. The aggregate data report the numbers of plants, openings, and closures by prefecture, industry, and survey wave, which we map to electricity regions. Because the data do not distinguish plant entry and exit from relocation across regions, we interpret these measures as capturing plant turnover.

A-2 Energy Consumption Statistics (ECS)

We combine the ECS with plant-level data from the Petroleum and Other Fuels Monthly Statistics (POFMS). The ECS is an annual survey covering a broad range of industries,

whereas POFMS is a monthly census of establishments in nine energy-intensive manufacturing industries: pulp, paper, and paperboard; chemicals; chemical fibers; petroleum products; ceramic, stone, and clay products; glass products; iron and steel; non-ferrous metal smelting and refining; and machinery. To avoid duplicate reporting, the ECS excludes establishments covered by POFMS. We aggregate the monthly POFMS observations to the fiscal-year level and harmonize the two datasets before pooling them.

The CMF, ECS, and POFMS all provide plant-level data but use different sampling frames. Our CMF sample covers manufacturing establishments with at least 30 employees. POFMS is a census of all establishments meeting its industry-specific eligibility criteria, which generally target large and energy-intensive plants. The ECS fully enumerates establishments on the self-generation roster, designated energy-management plants, and large establishments exceeding energy-use thresholds, while randomly sampling the remaining establishments by industry and size. Consequently, relative to the CMF, the pooled ECS–POFMS sample is more concentrated among large, energy-intensive, and self-generating plants. The surveys also have different response rates: in 2018, the response rates were 69.6% for ECS and 97.6% for POFMS, whereas the 2019 CMF achieved a response rate of 95.2%.

We construct total fuel consumption in common energy units in two steps since the surveys report a broad range of energy inputs, including electricity, gaseous fuels, petroleum products, coal and coke, and steam. First, we standardize each fuel's reported quantity. Most establishments report physical quantities, but some report monetary values; in these cases, we recover physical quantities by dividing expenditures by the corresponding regional fiscal-year fuel price. Because fuels are reported in different units, we express physical quantities in liters, kilograms, or cubic meters, as appropriate for each fuel. Second, we convert the resulting quantities into a common energy unit using conversion factors based primarily on the 2013 standard gross calorific values published by the ANRE (Agency for Natural Resources and Energy, 2022). We aggregate energy content across fuel types and divide by 1,000 to express total fuel use in GJ.

We construct fuel consumption for on-site generation using the reported purpose of fuel consumption. We define fuel used for self-generation as fuel consumed by boilers, cogeneration systems, and diesel generators. For each fuel type, we calculate fuel used directly in production as total fuel consumption minus fuel used by these self-generation facilities. Observations with negative constructed values are excluded, and the resulting measures are winsorized at the 1st and 99th percentiles.

The ECS does not directly report equipment ownership, so we infer self-generator ownership from observed self-generation. We classify a plant as owning a self-generator beginning in the first year it reports a strictly positive quantity of self-generated electricity; ownership persists thereafter. For the heterogeneity analysis by pre-accident ownership, plants are

classified as owners if they report positive self-generation by 2010.

A-3 Robustness-Control Variables

This appendix describes the regional variables used in the robustness checks in Section 5.5. We construct supply-chain exposure measures using prefecture-by-industry input-output tables from RIETI (RIETI, 2005). We define disaster-affected areas as Iwate, Miyagi, and Fukushima prefectures, which accounted for more than 99% of deaths and missing persons from the earthquake and tsunami. The direct exposure measure captures the share of a prefecture-industry's intermediate inputs sourced from disaster-affected areas. For example, if an iron factory sources 50% of its materials from disaster-affected areas, its direct exposure is 50%. The indirect exposure measure incorporates upstream input-output linkages to capture exposure through suppliers' suppliers. For example, a machinery plant that procures iron from suppliers using materials from disaster-affected areas would have indirect exposure. We use the 2005 tables, the latest available wave before the Fukushima accident. Because these exposure measures are time-invariant, the robustness specifications interact the pre-disaster measures with year indicators.

We construct local demand controls from prefecture-level economic indicators published by the Ministry of Internal Affairs and Communications (MIC, 2025). These controls include gross prefectural product in non-manufacturing sectors, total population, working-age population, construction investment, consumption, and the job-to-applicant ratio. Construction investment may be relevant because the earthquake increased reconstruction demand. The job-to-applicant ratio measures labor market tightness, which is especially relevant in the short run.

B Empirical Strategy

B-1 Measuring Electricity Supply Shocks

Table B1: Peak-Day Electricity Supply-Demand Conditions

	Demand 2010	Supply 2010	Planned Supply		
			2012	2014	2016
	(1)	(2)	(3)	(4)	(5)
A. Tohoku Region					
Total	15.6	16.6	14.8	15.5	15.1
Thermal	-	11.9	12.5	14.0	13.5
Nuclear	-	2.5	0.0	0.0	0.0
Hydro	-	1.9	1.4	1.7	1.6
Renewable	-	0.8	0.9	0.9	1.3
Transmission	-	-0.5	-0.1	-1.0	-1.2
Supply Shock	-	-	10.3%	5.3%	7.8%
B. Tokyo Region					
Total	59.9	64.1	57.7	56.1	52.0
Thermal	-	41.5	46.4	43.8	39.2
Nuclear	-	10.7	0.0	0.0	0.0
Hydro	-	3.4	3.0	3.0	2.7
Renewable	-	8.3	8.6	9.5	10.2
Transmission	-	0.3	-0.3	-0.2	-0.1
Supply Shock	-	-	8.8%	11.4%	18.3%
C. Chubu Region					
Total	27.1	29.9	27.9	27.4	27.4
Thermal	-	21.2	23.4	23.4	21.3
Nuclear	-	2.7	0.0	0.0	0.0
Hydro	-	1.5	1.4	1.4	1.4
Renewable	-	4.1	4.1	4.4	5.1
Transmission	-	0.3	-1.1	-1.9	-0.4
Supply Shock	-	-	2.2%	4.0%	3.9%
D. Kansai Region					
Total	31.0	32.7	27.8	29.6	27.4
Thermal	-	16.8	19.2	20.5	20.3
Nuclear	-	8.4	0.0	0.0	0.0
Hydro	-	2.3	2.5	2.6	2.6
Renewable	-	4.5	2.4	4.8	4.7
Transmission	-	0.7	1.2	1.7	-0.2
Supply Shock	-	-	15.2%	9.4%	16.4%

Notes: This table reports peak-day supply and demand (in GW) for four major regions. The peak day differs across regions. Transmission refers to electricity exchanged with utilities in other regions and with non-utility suppliers; thus, transmission does not necessarily sum to zero. Table constructed by the authors.

B-2 Derivation of the Small-Share Benchmark

Suppose P denotes the output price and P^K, P^L, P^M, and P^E denote input prices. A plant solves

$$\max_{K,L,M,E} \; P\Omega F(K,L,M,E) - P^K K - P^L L - P^M M - P^E E. \tag{B1}$$

Taking logs gives

$$\ln Y = \ln \Omega + \ln F(K,L,M,E). \tag{B2}$$

Differentiating with respect to shock severity $\mathcal{S}$ yields

$$\frac{d\ln Y}{d\mathcal{S}} = \frac{d\ln \Omega}{d\mathcal{S}} + \frac{1}{F}\frac{dF}{d\mathcal{S}}. \tag{B3}$$

By the chain rule,

$$\begin{aligned}\frac{1}{F}\frac{dF}{d\mathcal{S}} &= \frac{F_K}{F}\frac{dK}{d\mathcal{S}} + \frac{F_L}{F}\frac{dL}{d\mathcal{S}} + \frac{F_M}{F}\frac{dM}{d\mathcal{S}} + \frac{F_E}{F}\frac{dE}{d\mathcal{S}} \\ &= \frac{F_K K}{F}\frac{d\ln K}{d\mathcal{S}} + \frac{F_L L}{F}\frac{d\ln L}{d\mathcal{S}} + \frac{F_M M}{F}\frac{d\ln M}{d\mathcal{S}} + \frac{F_E E}{F}\frac{d\ln E}{d\mathcal{S}}.\end{aligned} \tag{B4}$$

Define the output elasticity with respect to each input as

$$\theta_K \equiv \frac{F_K K}{F}, \qquad \theta_L \equiv \frac{F_L L}{F}, \qquad \theta_M \equiv \frac{F_M M}{F}, \qquad \theta_E \equiv \frac{F_E E}{F}. \tag{B5}$$

Substituting these expressions into the derivative above yields

$$\frac{d\ln Y}{d\mathcal{S}} = \theta_E \frac{d\ln E}{d\mathcal{S}} + \theta_M \frac{d\ln M}{d\mathcal{S}} + \theta_L \frac{d\ln L}{d\mathcal{S}} + \theta_K \frac{d\ln K}{d\mathcal{S}} + \frac{d\ln \Omega}{d\mathcal{S}}. \tag{B6}$$

Since electricity is a flexible input, under price-taking behavior, the first-order condition for electricity is

$$P\Omega F_E = P^E. \tag{B7}$$

Multiplying this condition by electricity use and dividing by nominal revenue $PY = P\Omega F$ gives

$$\theta_E = \frac{F_E E}{F} = \frac{P^E E}{PY} = s_E. \tag{B8}$$

For the small-share benchmark, evaluate this condition at the pre-shock optimum, denoted by stars. Holding labor, materials, capital, and productivity fixed at L^*, M^*, K^*, and Ω^*, respectively, sets the non-electricity terms in equation (2) to zero. The direct first-order

effect of the shock is therefore

$$\left.\frac{d\ln Y}{d\mathcal{S}}\right|_{L=L^*,\,M=M^*,\,K=K^*,\,\Omega=\Omega^*} \approx \theta_E^* \frac{d\ln E}{d\mathcal{S}} = s_E^* \frac{d\ln E}{d\mathcal{S}}. \tag{B9}$$

The approximation is local and abstracts from induced adjustments in non-electricity inputs and productivity. This establishes equation (3).

B-3 Production Function Estimation

B-3.1 Derivation of the Flexible-Input Share Equation

The plant's static problem with respect to electricity input is

$$\max_{E_{it}} \; P_{it}\,\mathbb{E}\big[\exp\big(f(k_{it},l_{it},m_{it},e_{it})+\omega_{it}+\epsilon_{it}\big) \mid k_{it},l_{it},m_{it},e_{it},\omega_{it}\big] - P_t^E E_{it}, \tag{B10}$$

where $e_{it} \equiv \ln E_{it}$.

The first-order condition is

$$P_{it}\,\mathbb{E}\left[\frac{\partial Y_{it}}{\partial E_{it}} \mid k_{it},l_{it},m_{it},e_{it},\omega_{it}\right] = P_t^E. \tag{B11}$$

Using

$$\frac{\partial Y_{it}}{\partial E_{it}} = \frac{\partial f(k_{it},l_{it},m_{it},e_{it})}{\partial e_{it}} \frac{Y_{it}}{E_{it}}, \tag{B12}$$

we obtain

$$\frac{P_t^E E_{it}}{P_{it} Y_{it}} = \theta_{E,it}\,\mathbb{E}[\exp(\epsilon_{it}) \mid k_{it},l_{it},m_{it},e_{it}]\exp(-\epsilon_{it}), \tag{B13}$$

where

$$\theta_{E,it} \equiv \frac{\partial f(k_{it},l_{it},m_{it},e_{it})}{\partial e_{it}}. \tag{B14}$$

Imposing the normalization $\mathbb{E}[\exp(\epsilon_{it}) \mid k_{it},l_{it},m_{it},e_{it}] = 1$ yields

$$\ln s_{E,it} = \ln \theta_{E,it} - \epsilon_{it}. \tag{B15}$$

B-3.2 Recovering CES Distribution Parameters

Suppose the production aggregator within an industry is CES:

$$F(K,L,M,E) = \left(\sum_{j\in\{K,L,M,E\}} \alpha_j X_j^{\rho}\right)^{1/\rho}, \qquad \rho \equiv \frac{\sigma-1}{\sigma}, \tag{B16}$$

where $\sum_j \alpha_j = 1$ and σ is the elasticity of substitution across inputs. For $\sigma \neq 1$, the associated unit-cost function is proportional to

$$c(p) = \left[\sum_{j \in \{K,L,M,E\}} \alpha_j^{\sigma} p_j^{1-\sigma} \right]^{1/(1-\sigma)}. \tag{B17}$$

By Shephard's lemma, input j's share of total cost is

$$s_j^C(p) = \frac{\alpha_j^{\sigma} p_j^{1-\sigma}}{\sum_{k \in \{K,L,M,E\}} \alpha_k^{\sigma} p_k^{1-\sigma}}. \tag{B18}$$

For the CES technology above, cost minimization implies that an input's normalized output elasticity equals its cost share. We therefore construct the industry-specific cost-share target as

$$\hat{s}_j = \frac{\hat{\theta}_j}{\sum_{k \in \{K,L,M,E\}} \hat{\theta}_k}. \tag{B19}$$

Expressing input prices relative to their reference levels gives $p_j^0 = 1$ for every input. Equation (B18) then implies

$$\hat{s}_j = \frac{\hat{\alpha}_j(\sigma)^{\sigma}}{\sum_{k \in \{K,L,M,E\}} \hat{\alpha}_k(\sigma)^{\sigma}}. \tag{B20}$$

Conditional on a candidate σ and imposing $\sum_j \hat{\alpha}_j(\sigma) = 1$, the CES distribution parameters are therefore

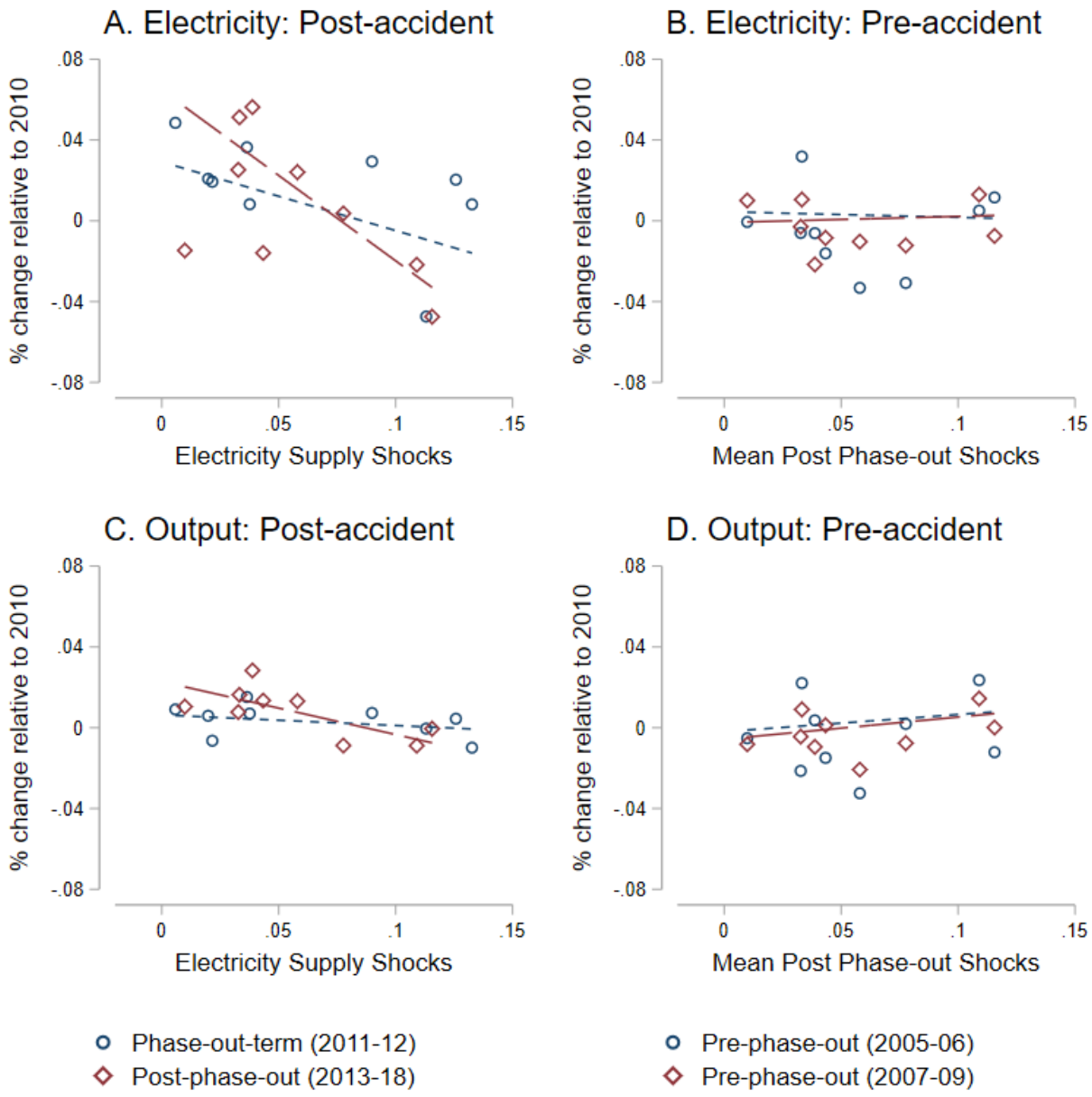
$$\hat{\alpha}_j(\sigma) = \frac{\hat{s}_j^{1/\sigma}}{\sum_{k \in \{K,L,M,E\}} \hat{s}_k^{1/\sigma}}, \qquad j \in \{K, L, M, E\}. \tag{B21}$$

Substituting Equation (B21) into the cost-share equation reproduces $\hat{s}_j$ for every candidate value of σ.

C Additional Results and Robustness Checks

C-1 Additional Results

Figure C1: Correlation between regional supply shock and power usage and revenue



Notes: We first take residuals in outcomes controlling for year-by-industry (3 digits) fixed effects. We then plot the differences in log residual relative to 2010 on the Y-axis and our measures of supply shock on the X-axis. Panel A and B use electricity, and Panels C and D use revenue as outcomes. Panels B and D use mean supply shocks during the post-phase-out period (2013-2018) on the x-axis.

Table C1: Effects of Supply Shocks on Labor

	Dep Var: $\Delta \ln(Y) = \ln(Y) - \ln(Y_{2010})$				
		Employment			
	Salary per worker (1)	Male (2)	Female (3)	Regular (4)	Non-regular (5)
Panel A: The effects of Supply Shocks using OLS					
Supply Shocks	0.01	-0.01	-0.05	-0.03	-0.21
(Phase-out Terms)	[0.71]	[0.77]	[0.46]	[0.64]	[0.26]
Supply Shocks	-0.04	-0.19**	-0.18**	-0.19	-0.43
(Post-Phase-out)	[0.51]	[0.01]	[0.02]	[0.11]	[0.16]
Panel B: The effects of Supply Shocks using 2SLS					
Supply Shocks	0.03	-0.02	-0.07*	-0.01	-0.40
(Phase-out Terms)	[0.34]	[0.61]	[0.08]	[0.79]	[0.10]
Supply Shocks	-0.01	-0.17**	-0.20**	-0.07	-0.90*
(Post-Phase-out)	[0.79]	[0.04]	[0.04]	[0.46]	[0.08]
Year-by-Industry FE	Yes	Yes	Yes	Yes	Yes

Notes: The table reports regression estimates. Columns (2) and (3) refer to male and female employment, respectively. Columns (4) and (5) refer to regular and non-regular employment. The "Phase-out Terms" row uses data from 2005–2010 and 2011–2012. The "Post-Phase-out" row uses data from 2005–2010 and 2013–2018. All regressions include year-by-industry (3-digit) fixed effects and use 2010 log-salary weights. We use wild cluster bootstrap with regions as clusters. P-values are reported in brackets. $^{***}p < 0.01, ^{**}p < 0.05, ^{*}p < 0.1$.

C-2 Electricity-Biased Technical Change

We ask whether the post-accident decline in electricity intensity reflects (i) cost-minimizing factor substitution driven by relative input prices, or (ii) a shift in electricity-augmenting technology (directed technical change). Under a CES aggregator over electricity and labor, relative input demand can be written as

$$\ln\left(\frac{E_{it}}{L_{it}}\right) = -\sigma \ln\left(\frac{p_{it}^{E}}{p_{it}^{L}}\right) + (1-\sigma)\,\omega_{it}^{E}, \tag{C1}$$

where σ is the elasticity of substitution and ω_{it}^{E} captures non-price shifts in electricity-augmenting technology.[2] Increases in p^E/p^L reduce E/L, while changes in ω_{it}^{E} shift E/L beyond what prices predict.

Recent manufacturing estimates are approximately 0.1 for labor–electricity substitution in India (Aggarwal, 2026), 0.24–0.26 in the United States (Hawkins-Pierot and Wagner, 2026), and 0.5–0.55 for broader energy inputs in Indonesia (Abe et al., 2026).

We assess whether relative price movements account for the observed evolution of E/L after Fukushima. Rather than estimate σ from potentially endogenous relative prices, we consider $\sigma \in \{0.2, 0.5, 0.7\}$. For each value, we recover the change in electricity-biased technology as

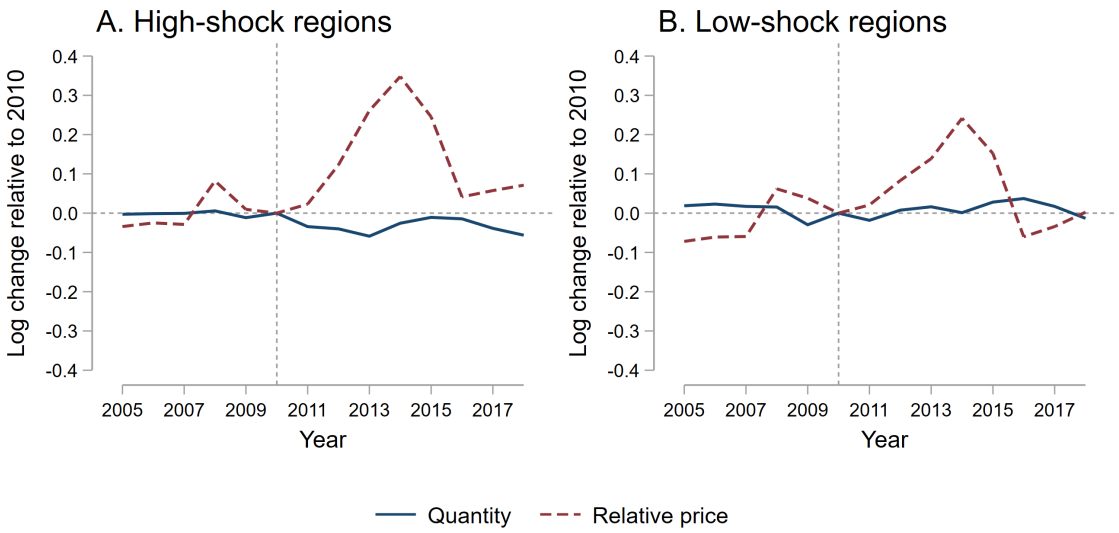
$$\Delta\omega_{it}^{E}(\sigma) = \frac{\Delta \ln(E_{it}/L_{it}) + \sigma \Delta \ln(p_{it}^{E}/p_{it}^{L})}{1-\sigma}. \tag{C2}$$

This measure captures changes in electricity productivity relative to labor productivity under the CES framework.

Appendix Figure C2 plots the relative quantity and price series for high- and low-shock regions. Relative prices rose more and the electricity-to-labor ratio fell more in high-shock regions (Panel A) than in low-shock regions (Panel B), consistent with substitution away from electricity. Appendix Table C2 reports OLS and 2SLS estimates of the effect of supply shocks on implied electricity-biased productivity. The post-phase-out estimates are statistically insignificant across all assumed elasticities ($\sigma \in \{0.2, 0.5, 0.7\}$). Depending on the assumed elasticity, the coefficient estimates change sign and range from -0.6 to 0.3. Overall, we find little evidence that supply shocks induced electricity-biased technical change.

[2] Equation (C1) follows from CES demand with factor-augmenting terms: changes in p^E/p^L move the plant along an isoquant, while changes in ω^E shift relative demand holding prices fixed.

Figure C2: Relative Electricity Inputs and Prices



Notes: The figure plots changes in the log electricity-to-labor ratio and the log relative price of electricity to labor. Panel A displays regions with supply shocks above the median ("High-shock"), while Panel B displays those below the median ("Low-shock"). All variables are log-differences relative to 2010.

Table C2: Effects of Supply Shocks on Electricity-Biased Technical Change

	Assumed Elasticity of Substitution		
	$\sigma = 0.2$	$\sigma = 0.5$	$\sigma = 0.7$
	(1)	(2)	(3)
Panel A. Effects of supply shocks using OLS			
Supply shocks	-0.26**	-0.41**	-0.66**
(Phase-out Terms)	[0.05]	[0.03]	[0.03]
Supply shocks	-0.38*	-0.13	0.33
(Post-Phase-out)	[0.06]	[0.69]	[0.54]
Panel B. Effects of supply shocks using 2SLS			
Supply shocks	-0.23	-0.38**	-0.65*
(Phase-out Terms)	[0.21]	[0.05]	[0.08]
Supply shocks	-0.60	-0.27	0.30
(Post-Phase-out)	[0.31]	[0.85]	[0.77]
Year-by-industry FE	Yes	Yes	Yes

Notes: The dependent variable is the implied change in electricity-biased technology, $\Delta\omega^E(\sigma) = [\Delta \ln(E/L) + \sigma\Delta \ln(p^E/p^L)]/(1-\sigma)$. Columns (1)–(3) assume elasticities of substitution of 0.2, 0.5, and 0.7, respectively. The phase-out rows use data from 2005–2010 and 2011–2012. The post-phase-out rows use data from 2005–2010 and 2013–2018. All regressions include year-by-industry fixed effects, where industries are defined at the three-digit level, and use 2010 log-labor weights. The 2SLS specifications instrument supply shocks with predicted changes in nuclear generation and the initial reserve margin. P-values from a wild cluster bootstrap using regions as clusters are reported in brackets. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

C-3 Alternative Explanations and Robustness Checks

Table C3: Supply-Chain Exposure and Local-Demand Controls

	(1)	(2)	(3)	(4)	(5)
	Dep Var: $\Delta \ln(E) = \ln(E) - \ln(E_{2010})$				
Panel A: The effects of Supply Shocks using OLS					
Supply Shocks	-0.24	-0.24*	-0.24*	-0.24	-0.22
(Phase-out Terms)	[0.11]	[0.10]	[0.10]	[0.11]	[0.19]
Supply Shocks	-0.63**	-0.68**	-0.69***	-0.69***	-0.55***
(Post-Phase-out)	[0.03]	[0.01]	[0.01]	[0.01]	[0.01]
Panel B: The effects of Supply Shocks using 2SLS					
Supply Shocks	-0.21	-0.29	-0.28	-0.28	-0.28
(Phase-out Terms)	[0.36]	[0.16]	[0.16]	[0.16]	[0.28]
Supply Shocks	-0.89*	-0.87**	-0.87**	-0.88**	-0.75**
(Post-Phase-out)	[0.06]	[0.02]	[0.01]	[0.01]	[0.02]
	Dep Var: $\Delta \ln(R) = \ln(R) - \ln(R_{2010})$				
Panel C: The effects of Supply Shocks using OLS					
Supply Shocks	-0.04	-0.01	-0.01	-0.01	-0.02
(Phase-out Terms)	[0.23]	[0.72]	[0.75]	[0.81]	[0.54]
Supply Shocks	-0.19***	-0.18**	-0.18**	-0.18*	-0.18
(Post-Phase-out)	[0.00]	[0.05]	[0.05]	[0.06]	[0.11]
Panel D: The effects of Supply Shocks using 2SLS					
Supply Shocks	-0.02	0.00	0.00	0.00	0.00
(Phase-out Terms)	[0.61]	[0.95]	[0.96]	[0.95]	[0.98]
Supply Shocks	-0.23**	-0.20**	-0.20**	-0.20*	-0.23
(Post-Phase-out)	[0.04]	[0.05]	[0.05]	[0.06]	[0.12]
Direct supply-chain	No	Yes	No	Yes	No
Indirect supply-chain	No	No	Yes	Yes	No
Local-demand	No	No	No	No	Yes
Year-by-Industry FE	Yes	Yes	Yes	Yes	Yes

Notes: Column (1) reproduces the baseline estimates in Table 4 using the full analysis sample. Columns (2)–(5) exclude plants in Iwate, Miyagi, and Fukushima and use a common complete-case sample. Direct exposure measures the share of intermediate inputs sourced from these prefectures. Indirect exposure incorporates upstream linkages through the Leontief inverse. Supply-chain exposure measures are interacted with year indicators. Local-demand controls include gross prefectural product in non-manufacturing sectors, total and working-age populations, construction investment, consumption, and the job-to-applicant ratio. See Appendix A-3 for variable construction. All regressions include three-digit-industry-by-year fixed effects and use 2010 log-labor weights. P-values from a wild cluster bootstrap using regions as clusters are reported in brackets. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table C4: Robustness Checks

	Plant Controls (1)	Two-way FE (2)	No Weights (3)
	Electricity Use		
Panel A: The effects of Supply Shocks using OLS			
Supply Shocks	-0.26*	-0.17	-0.23
(Phase-out Terms)	[0.08]	[0.23]	[0.12]
Supply Shocks	-0.64**	-0.48**	-0.62**
(Post-Phase-out)	[0.03]	[0.02]	[0.03]
Panel B: The effects of Supply Shocks using 2SLS			
Supply Shocks	-0.32	-0.21	-0.29
(Phase-out Terms)	[0.22]	[0.47]	[0.21]
Supply Shocks	-0.82**	-0.79***	-0.78**
(Post-Phase-out)	[0.02]	[0.01]	[0.02]
	Output		
Panel C: The effects of Supply Shocks using OLS			
Supply Shocks	-0.06	-0.03	-0.04
(Phase-out Terms)	[0.36]	[0.44]	[0.19]
Supply Shocks	-0.16	-0.14	-0.19***
(Post-Phase-out)	[0.24]	[0.17]	[0.00]
Panel D: The effects of Supply Shocks using 2SLS			
Supply Shocks	-0.06	0.01	-0.04
(Phase-out Terms)	[0.39]	[0.94]	[0.33]
Supply Shocks	-0.20	-0.17	-0.24***
(Post-Phase-out)	[0.19]	[0.35]	[0.00]
Year-by-Industry FE	Yes	Yes	Yes

Notes: Columns (1)–(3) add plant controls, use two-way fixed effects, and omit the baseline weights, respectively. All regressions include three-digit-industry-by-year fixed effects. See the main text for details on each robustness check. We use a wild cluster bootstrap to obtain p-values, clustering by region. P-values are reported in brackets. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

D Plants' Responses to Supply Shocks

Table D1: Spillover Effects: Production Reallocation Across Regions

Dep Var: $\Delta \ln(Y) = \ln(Y) - \ln(Y_{2010})$	Electricity (1)	Output (2)
Panel A: The effects of Supply Shocks using OLS		
Supply Shocks	0.24	-0.02
(Phase-out Terms)	[0.16]	[0.70]
Supply Shocks	0.48**	-0.06
(Post-Phase-out)	[0.04]	[0.59]
Panel B: The effects of Supply Shocks using 2SLS		
Supply Shocks	0.19	-0.03
(Phase-out Terms)	[0.44]	[0.74]
Supply Shocks	0.84**	-0.04
(Post-Phase-out)	[0.04]	[0.80]
Year-by-Industry FE	Yes	Yes

Notes: The dependent variables are log changes in outcomes for the firm's plants in all other regions (excluding the focal region), measured relative to 2010. The regressor is the supply shock in the focal region. A positive coefficient indicates that a negative shock in the focal region is associated with increased activity elsewhere within the firm. All the regressions include year-by-industry (3-digits) fixed effects and use 2010 log-salary weights. We use wild cluster bootstrap using regions as clusters. P-values are reported in brackets. $^{***}p < 0.01,^{**} p < 0.05,^{*} p < 0.1$.

Table D2: Effects of Supply Shocks on Entry and Exit

	Number Plants (1)	Entry (2)	Exit (3)
Panel A: The effects of Supply Shocks using OLS			
Supply Shocks	-0.30	0.74	0.44
	[0.18]	[0.56]	[0.21]
Panel B: The effects of Supply Shocks using 2SLS			
Supply Shocks	0.09	-0.10	0.18
	[0.82]	[0.94]	[0.75]
Year-by-Industry FE	Yes	Yes	Yes

Notes: The table reports estimates of equation (5) for plant counts and turnover outcomes. The dependent variables are log differences relative to the 2009 (pre-Fukushima) wave. The sample includes the 2009, 2014, and 2016 waves. P-values, reported in brackets, are obtained from a wild cluster bootstrap clustered at the region level. $^{***}p < 0.01,^{**}p < 0.05,^{*}p < 0.1$.

D-1 Electricity Supply Shocks and On-Site Generation

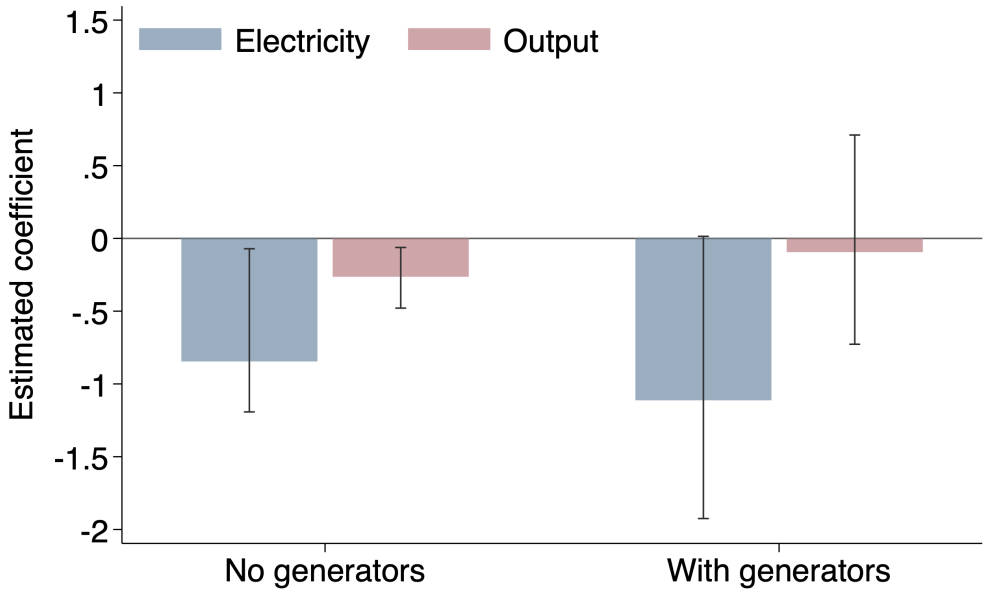


Figure D1: Effects of Supply Shocks by Generator Ownership

Notes: The figure reports post-phase-out 2SLS estimates for grid electricity purchases and output, estimated separately for plants with and without on-site generators. Generator owners are plants that reported positive self-generation by 2010; the non-generator group consists of the remaining matched plants. The regressions use data from 2005–2010 and 2013–2018, include three-digit-industry-by-year fixed effects, and use 2010 log-salary weights. The vertical bars show 90% confidence intervals obtained from a wild cluster bootstrap using regions as clusters.

E From Reduced-Form Effects to Production Mechanisms

E-1 Simulation Inputs

Table E1 summarizes the technology, demand, and policy inputs used in the simulation. Three features of the scenario construction are worth noting. First, we hold technology and demand parameters fixed between the no-shock baseline and the post-phase-out scenario. Second, we express all input prices relative to their 2010 levels. Relative non-electricity prices remain one, while the two scenarios differ in the relative electricity price and shock-season rationing.

Table E1: Simulation Inputs

Object	Level	Description or source	Baseline	Post-phase-out
Panel A: Technology				
σ	Common	Elasticity of substitution	$\widehat{\sigma}$	$\widehat{\sigma}$
$\widehat{\boldsymbol{\theta}}_j$	Industry	Estimated output elasticities	$\widehat{\boldsymbol{\theta}}_j$	$\widehat{\boldsymbol{\theta}}_j$
Ω_i	Plant	Baseline productivity	Ω_i	Ω_i
κ	Common	Capital-utilization elasticity	$\widehat{\kappa}$	$\widehat{\kappa}$
v_i	Plant	Baseline capital–electricity ratio; $\frac{K_{2010}}{E_{2010}}$	v_i	v_i
G_i	Plant	Generator ownership	G_i	G_i
Panel B: Demand				
η	Common	Demand elasticity; 5	5	5
B_i	Plant	Demand shifter; $Y_{i,2010}$	B_i	B_i
Panel C: Policies				
R_r	Region	Shock-season electricity-saving target	0	$\overline{R}_r$
ϕ	Common	Shock-season share of the year	-	90/365
p_r^E	Region	Regional electricity price	1	$\overline{p}_r^E$
p^K, p^L, p^M	Common	Non-electricity input prices	1	1

E-2 Predicted Electricity-Price Changes

To construct the shock-induced electricity-price input used in the simulation, we estimate how regional electricity supply shocks affected the prices paid by manufacturing plants. Observed electricity prices increased through 2014 and declined during 2015–2016, largely reflecting nationwide movements in global energy costs. We therefore control for nationwide electricity-price trends to isolate the component attributable to regional supply shocks. We estimate the following specification separately for the phase-out period (2011–2012) and the

post-phase-out period (2013–2018):

$$\Delta \ln p_{rt}^{E} \equiv \ln p_{rt}^{E} - \ln p_{r,2010}^{E} = \beta\, \mathcal{S}_{rt} + \gamma_{jt} + \varepsilon_{ijrt}, \tag{E1}$$

where p_{rt}^{E} denotes the average electricity price in region r and year t, which we assign to plant i; $\mathcal{S}_{rt}$ denotes the region–year supply shock; and γ_{jt} denotes three-digit-industry-by-year fixed effects. These fixed effects absorb nationwide price movements, including common changes in fossil-fuel costs. We estimate Equation (E1) by OLS and 2SLS, using the instruments defined in Equation (7) for the latter. All specifications use the same 2010 total-salary weights as the main analysis, and standard errors are clustered by region.

Table E2 shows no detectable price response during 2011–2012: both the OLS and 2SLS estimates are close to zero. This pattern is consistent with adjustment lags in regulated tariffs under Japan's cost-of-service regulation. During 2013–2018, the OLS and 2SLS estimates are 0.78 and 1.12, respectively. The 2SLS estimate implies that a 1 ppt increase in the supply shock raises the log electricity price by 0.0112, corresponding to an approximately 1.12% increase.

For the simulation, we normalize the 2010 electricity price to one and use the post-phase-out 2SLS estimate to construct each region's predicted price:

$$\overline{p}_{r}^{E} \equiv \frac{p_{r,\text{post}}^{E}}{p_{r,2010}^{E}} = \exp\!\left(\widehat{\beta}_{\text{post}}\, \overline{\mathcal{S}}_{r}\right), \tag{E2}$$

where $\overline{\mathcal{S}}_{r}$ is the region's mean supply shock over 2013–2018. The sample's weighted mean shock is 7.4%. The corresponding weighted mean predicted log-price change is 8.4 log points, while the weighted mean of predicted relative prices implies an approximately 9.0% increase in levels. In Tokyo, where the mean shock is 10.8%, the predicted log-price change is 12.9 log points, corresponding to an approximately 13.8% increase in levels. Because regulated tariffs do not vary seasonally within a year, we apply the same predicted price during the shock and off-shock seasons.

Table E2: Effects of Electricity Supply Shocks on Electricity Prices

	Dep Var: $\Delta \ln p^E = \ln p^E_{rt} - \ln p^E_{r,2010}$	
	Phase-out	Post-phase-out
	(1)	(2)
Panel A. The effects of supply shocks using OLS		
Supply shock	0.03	0.78***
	(0.19)	(0.20)
Panel B. The effects of supply shocks using 2SLS		
Supply shock	0.00	1.12**
	(0.32)	(0.35)
Year-by-Industry FE	Yes	Yes

Notes: Column (1) uses data from 2005–2010 and 2011–2012. Column (2) uses data from 2005–2010 and 2013–2018. Panel B instruments the supply shock with predicted nuclear-supply losses and the 2010 reserve margin. All regressions include three-digit-industry-by-year fixed effects. Standard errors clustered by region are in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

E-3 Predictions under the Standard CES Technology

We first predict inputs and output under the electricity-supply-shock scenario using the standard CES technology. In this subsection, we abstract from electricity-induced capital idleness by setting $\kappa = 0$. We introduce electricity-dependent capital utilization in Appendix E-4.

E-3.1 Baseline and price-only equilibria

The simulation considers alternative values of the elasticity of substitution, σ. For each candidate value, we recover the CES distribution parameters α using Equation (15), while holding the baseline cost shares in Equation (14) fixed.

Plant i in industry j chooses inputs, output, and its output price to maximize profits:

$$\begin{aligned} \max_{K,L,M,E,P,Y} \quad & \Pi_i = P_i Y_i - p^K K_i - p^L L_i - p^M M_i - p^E E_i, \\ \text{subject to} \quad & Y_i = \Omega_i \left(\alpha_{Kj} K_i^\rho + \alpha_{Lj} L_i^\rho + \alpha_{Mj} M_i^\rho + \alpha_{Ej} E_i^\rho\right)^{1/\rho}, \\ & Y_i = B_i P_i^{-\eta}, \qquad \rho = \frac{\sigma - 1}{\sigma}. \end{aligned} \tag{E3}$$

For $\sigma \neq 1$, substitution of Equation (15) into the dual cost function gives

$$c_{ij}(p) = \frac{1}{\Omega_i} \left[\sum_{h \in \{K,L,M,E\}} s_{hj} (p^h)^{1-\sigma} \right]^{1/(1-\sigma)}, \tag{E4}$$

where each p^h is an input-price index relative to its 2010 level. Hence $p^{h,0} = 1$ and $c_{ij}(p^0) = 1/\Omega_i$ for every value of σ. When only the electricity price changes, the relative unit-cost index is

$$\frac{c_{ij}(p^1)}{c_{ij}(p^0)} = \left[1 - s_{Ej} + s_{Ej}(p^{E,1})^{1-\sigma}\right]^{1/(1-\sigma)}. \tag{E5}$$

This expression makes clear that comparisons across σ hold the baseline electricity share fixed. Constant-elasticity demand implies the markup $\mu = \eta/(\eta - 1)$, so the optimal output price and output are

$$P_i^*(p) = \mu c_{ij}(p), \qquad Y_i^*(p) = B_i[\mu c_{ij}(p)]^{-\eta}. \tag{E6}$$

The corresponding conditional factor demands are

$$X_{ih}^*(p) = \frac{Y_i^*(p)}{\Omega_i}\left(\frac{c_{ij}(p)\Omega_i\alpha_{hj}}{p^h}\right)^{\sigma}, \qquad h \in \{K, L, M, E\}. \tag{E7}$$

We express each simulated outcome as the same log difference from the no-shock baseline used in the empirical analysis. Let $p^0 = (1, 1, 1, 1)$ denote the vector of 2010-relative price indices, and let p^1 denote the post-phase-out relative-price vector. Define

$$\Delta \ln c_{ij} \equiv \ln c_{ij}(p^1) - \ln c_{ij}(p^0), \qquad \Delta \ln p^h \equiv \ln p^{h,1} - \ln p^{h,0}. \tag{E8}$$

Because productivity, demand, and the markup are held fixed across simulations, they cancel from the log differences. Taking logs of Equations (E6) and (E7) gives

$$\begin{aligned}
\Delta \ln Y_i &= -\eta\, \Delta \ln c_{ij}, \\
\Delta \ln V_i &= (1 - \eta)\, \Delta \ln c_{ij}, \\
\Delta \ln X_{ih} &= (\sigma - \eta)\, \Delta \ln c_{ij} - \sigma\, \Delta \ln p^h, \qquad h \in \{K, L, M, E\}.
\end{aligned} \tag{E9}$$

Here, $V_i \equiv P_i Y_i$ denotes revenue and $\Delta \ln Z_i \equiv \ln Z_i^1 - \ln Z_i^0$ for any outcome Z_i. Thus, we predict the plant-level log changes used in the empirical analysis without using observed plant-level baseline inputs.

For the Cobb–Douglas benchmark, $\sigma = 1$, we evaluate these expressions at the continuous limit of the CES cost function. In this case,

$$\Delta \ln c_{ij} = \sum_{h \in \{K,L,M,E\}} s_{hj}\, \Delta \ln p^h. \tag{E10}$$

E-3.2 Electricity-price elasticities

Define electricity's share in unit cost as

$$s_{E,ij}^{C}(p) = \frac{s_{Ej}(p^{E})^{1-\sigma}}{\sum_{h\in\{K,L,M,E\}} s_{hj}(p^{h})^{1-\sigma}}. \tag{E11}$$

At the baseline relative-price vector $p^{0} = (1, 1, 1, 1)$, this expression equals s_{Ej} for every σ. By Shephard's Lemma, $\partial \ln c_{ij}(p)/\partial \ln p^{E} = s_{E,ij}^{C}(p)$. Combining this result with demand and the conditional factor demands gives

$$\begin{aligned}
\frac{d\ln Y_i}{d\ln p^{E}} &= -\eta s_{E,ij}^{C}, \\
\frac{d\ln X_{ih}}{d\ln p^{E}} &= (\sigma-\eta)s_{E,ij}^{C}, \qquad h\in\{K,L,M\}, \\
\frac{d\ln E_i}{d\ln p^{E}} &= -\sigma + (\sigma-\eta)s_{E,ij}^{C}.
\end{aligned} \tag{E12}$$

Producer surplus is $\Pi_i = (P_i - c_{ij})Y_i = (\mu-1)c_{ij}Y_i$, implying

$$\frac{d\ln \Pi_i}{d\ln p^{E}} = (1-\eta)s_{E,ij}^{C}. \tag{E13}$$

E-3.3 Seasonal rationing

Rationing applies only during the shock season S. For a plant in region r, shock-season electricity use must satisfy

$$E_i^{S} \le \overline{E}_i^{S} \equiv (1-\overline{R}_r)E_i^{S,0}, \tag{E14}$$

where $E_i^{S,0}$ denotes baseline shock-season electricity use. We first solve the unconstrained equilibrium under post-phase-out prices and denote the resulting price-only electricity demand by $E_i^{S,\mathrm{ref}}$.

If $E_i^{S,\mathrm{ref}} \le \overline{E}_i^{S}$, rationing is slack and the price-only solution applies. If $E_i^{S,\mathrm{ref}} > \overline{E}_i^{S}$, we represent the binding constraint with an incremental shadow price. Define

$$\Delta\ln \widetilde{p}_i^{E} \equiv \ln\left(\frac{\widetilde{p}_i^{E}}{p_r^{E,1}}\right), \qquad \Delta\ln E_i^{S,R} \equiv \ln\left(\frac{\overline{E}_i^{S}}{E_i^{S,\mathrm{ref}}}\right). \tag{E15}$$

Using a first-order approximation around the unconstrained post-price equilibrium, the shadow-price adjustment is

$$\Delta\ln \widetilde{p}_i^{E} = \frac{\Delta\ln E_i^{S,R}}{-\sigma + (\sigma-\eta)s_{E,ij}^{C}}, \tag{E16}$$

where the electricity cost share is evaluated at the unconstrained post-price equilibrium. The corresponding additional rationing-induced changes are

$$\begin{aligned}
\Delta \ln Y_i^{S,R} &= -\eta s_{E,ij}^{C} \Delta \ln \widetilde{p}_i^{E}, \\
\Delta \ln X_{ih}^{S,R} &= (\sigma - \eta) s_{E,ij}^{C} \Delta \ln \widetilde{p}_i^{E}, \qquad h \in \{K, L, M\}, \\
\Delta \ln \Pi_i^{S,R} &= (1 - \eta) s_{E,ij}^{C} \Delta \ln \widetilde{p}_i^{E}.
\end{aligned} \tag{E17}$$

The total shock-season response equals the price-only response plus this rationing adjustment. During the off-season O, plants face post-phase-out prices but no rationing.

Annual aggregation. We combine the seasonal predictions using the shock-season share $\phi = 90/365$:

$$\Delta \ln Z_i \equiv \phi \, \Delta \ln Z_i^{S} + (1 - \phi) \, \Delta \ln Z_i^{O}. \tag{E18}$$

This expression is a first-order approximation to the log change in an annual outcome constructed by aggregating the two seasons in levels.

E-3.4 Plants with generators

For plants that own generators, we retain the model-simulated response of grid-purchased electricity but set the structural responses of capital, labor, materials, output, and revenue to zero. This treatment is motivated by the heterogeneous reduced-form evidence in Section 6.2: generator owners reduce grid purchases, consistent with substitution toward self-generation, but exhibit no meaningful declines in non-electricity inputs or output. The ownership adjustment is imposed after the CES solution and is not implied by the first-order conditions above.

E-4 Incorporating Capital Idleness Induced by Electricity Supply Shocks

We use effective capital services, rather than installed capital, to predict inputs and output under electricity supply shocks. Effective capital services are defined in Equation (19) as

$$\widetilde{K}_{it} = \left[(1 - \kappa) K_{it}^{\rho} + \kappa (v_i E_{it})^{\rho} \right]^{1/\rho}, \qquad v_i = \frac{K_{i,2010}}{E_{i,2010}}.$$

Because $v_i E_{i,2010} = K_{i,2010}$, effective and installed capital coincide at the 2010 baseline: $\widetilde{K}_{i,2010} = K_{i,2010}$. Moreover, the baseline elasticities of effective capital services with respect

to installed capital and electricity are

$$\left.\frac{\partial \ln \widetilde{K}_{it}}{\partial \ln K_{it}}\right|_{t=2010} = 1-\kappa, \qquad \left.\frac{\partial \ln \widetilde{K}_{it}}{\partial \ln E_{it}}\right|_{t=2010} = \kappa. \tag{E19}$$

To incorporate capital idleness, we adjust the estimated capital and electricity output elasticities. Let $\widehat{\theta}_{Kj}$ denote the estimated capital elasticity. Equation (E19) shows that, at the 2010 baseline, a 1% change in installed capital changes effective capital services by $(1-\kappa)$%. Using the chain rule, the adjusted capital elasticity ($\widetilde{\theta}_{Kj}(\kappa)$) is

$$\begin{aligned}\widetilde{\theta}_{Kj}(\kappa) &\equiv \left.\frac{\partial \ln Y_{it}}{\partial \ln K_{it}}\right|_{t=2010} \\ &= \left.\frac{\partial \ln Y_{it}}{\partial \ln \widetilde{K}_{it}}\frac{\partial \ln \widetilde{K}_{it}}{\partial \ln K_{it}}\right|_{t=2010} \\ &= (1-\kappa)\widehat{\theta}_{Kj}. \end{aligned} \tag{E20}$$

Electricity affects output directly and through capital idleness. At the 2010 baseline, a 1% change in electricity changes effective capital services by κ%. The adjusted electricity elasticity ($\widetilde{\theta}_{Ej}(\kappa)$) therefore equals the original electricity elasticity plus κ times the capital elasticity:

$$\begin{aligned}\widetilde{\theta}_{Ej}(\kappa) &\equiv \left.\frac{d \ln Y_{it}}{d \ln E_{it}}\right|_{t=2010} \\ &= \widehat{\theta}_{Ej} + \left.\frac{\partial \ln Y_{it}}{\partial \ln \widetilde{K}_{it}}\frac{\partial \ln \widetilde{K}_{it}}{\partial \ln E_{it}}\right|_{t=2010} \\ &= \widehat{\theta}_{Ej} + \kappa\widehat{\theta}_{Kj}. \end{aligned} \tag{E21}$$

The output elasticities of labor and materials remain unchanged:

$$\widetilde{\theta}_{Lj}(\kappa) = \widehat{\theta}_{Lj}, \qquad \widetilde{\theta}_{Mj}(\kappa) = \widehat{\theta}_{Mj}. \tag{E22}$$

This adjustment reallocates part of the original capital elasticity to electricity but does not change the sum of the output elasticities.

Using these adjusted output elasticities, we recalculate the industry-specific baseline cost shares as

$$\widetilde{s}_{hj}(\kappa) \equiv \frac{\widetilde{\theta}_{hj}(\kappa)}{\sum_{g\in\{K,L,M,E\}} \widetilde{\theta}_{gj}(\kappa)}, \qquad h \in \{K,L,M,E\}. \tag{E23}$$

Because the sum of the output elasticities is unchanged, these shares can equivalently be

written as

$$\widetilde{s}_{Kj}(\kappa) = (1-\kappa)s_{Kj}, \qquad \widetilde{s}_{Lj}(\kappa) = s_{Lj},$$
$$\widetilde{s}_{Mj}(\kappa) = s_{Mj}, \qquad \widetilde{s}_{Ej}(\kappa) = s_{Ej} + \kappa s_{Kj}. \tag{E24}$$

Finally, for each candidate value of σ, we recover the CES distribution parameters from these adjusted baseline shares following Equation (15). Thus, κ changes the baseline allocation of the capital and electricity shares.

E-4.1 Minimum-Distance Estimation

We jointly estimate the elasticity of substitution, σ, and the capital-utilization elasticity, κ, by minimizing the distance between the model-implied changes and their reduced-form counterparts. We first construct the normalized baseline shares in Equation (14). For each candidate κ, we form the effective shares in Equation (E23); for each candidate σ, we map those fixed shares into CES coefficients using Equation (15). We then obtain annual plant-level changes in electricity, labor, materials, and output using the simulation procedure described in Appendix E-3. We aggregate these predictions using the same baseline salary weights as in the main reduced-form regressions. For each outcome, the empirical target is the post-phase-out 2SLS coefficient multiplied by the weighted mean supply shock. The moment discrepancies are

$$g_q(\sigma,\kappa) \equiv \frac{\sum_i w_i \Delta \ln q_i^{\text{sim}}(\sigma,\kappa)}{\sum_i w_i} - \widehat{\beta}_q^{\text{RF}} \overline{\mathcal{S}}_w, \qquad q \in \{E, L, M, Y\}, \tag{E25}$$

where w_i denotes the corresponding regression weight and $\overline{\mathcal{S}}_w \equiv \sum_i w_i \mathcal{S}_i / \sum_i w_i$ is the weighted mean supply shock.

Using an identity weighting matrix, we define the minimum-distance criterion as

$$Q(\sigma,\kappa) = \sum_{q \in \{E,L,M,Y\}} g_q(\sigma,\kappa)^2. \tag{E26}$$

The estimator is

$$(\widehat{\sigma}, \widehat{\kappa}) = \underset{\sigma>0,\, 0\le\kappa\le 1}{\arg\min}\; Q(\sigma,\kappa). \tag{E27}$$

We first evaluate $Q(\sigma,\kappa)$ over a coarsely spaced grid of candidate parameter values to identify the region containing the minimum and then evaluate it over a more finely spaced grid within that region to obtain $(\widehat{\sigma}, \widehat{\kappa})$.

E-4.2 Nested CES Robustness

The baseline model uses a common substitution elasticity for the effective-capital composite and the remaining inputs. We examine whether the results change when electricity and capital services have a separate, lower within-nest elasticity. Let σ_{EK} denote the elasticity between installed capital and the electricity services used to operate it, and let σ_O denote the outer elasticity between the resulting composite, labor, materials, and direct electricity services. Holding the baseline inner shares fixed, the price index for the electricity–capital composite is

$$c_{EK}(p^E; \sigma_{EK}, \kappa) = \left[(1-\kappa) + \kappa (p^E)^{1-\sigma_{EK}}\right]^{1/(1-\sigma_{EK})}. \tag{E28}$$

For $\sigma_{EK} = 0$, we evaluate Equation (E28) at its Leontief limit,

$$c_{EK}(p^E; 0, \kappa) = (1-\kappa) + \kappa p^E.$$

This dual formulation avoids evaluating the numerically unstable primal CES exponent when σ_{EK} is close to zero.

Using the normalized production-function elasticities as baseline outer shares, the outer unit-cost index is

$$c_j(p^E) = \left[s_{Kj} c_{EK}(p^E)^{1-\sigma_O} + s_{Lj} + s_{Mj} + s_{Ej} (p^E)^{1-\sigma_O}\right]^{1/(1-\sigma_O)}. \tag{E29}$$

Thus, changing either elasticity changes the curvature of the technology but not its baseline shares. For each imposed value $\sigma_{EK} \in \{0, 0.10, 0.30, 0.50, 0.70\}$, we recover the corresponding inner and outer CES coefficients and re-estimate (σ_O, κ) using the minimum-distance moments in Equation (E25), subject to $\sigma_O \leq 1$. We focus on this constrained calibration because the nested specification treats the electricity–capital composite as weakly complementary to labor and materials.

Table E3: Nested CES Calibration by Within-Nest Elasticity

	Outer complementarity: $\sigma_O \leq 1$		Predicted changes				RMSE
σ_{EK}	$\widehat{\sigma}_O$	$\widehat{\kappa}$	$\Delta \ln E$	$\Delta \ln L$	$\Delta \ln M$	$\Delta \ln Y$	
0.00	1.00†	0.20	-6.10	-1.47	-1.47	-1.84	0.344
0.10	1.00†	0.19	-6.42	-1.39	-1.39	-1.73	0.168
0.30	0.88	0.18	-6.74	-1.35	-1.35	-1.63	0.058
0.50	0.66	0.17	-6.75	-1.36	-1.36	-1.57	0.090
0.70	0.45	0.16	-6.75	-1.38	-1.38	-1.51	0.120
Reduced-form target			-6.72	-1.29	-1.29	-1.72	

Notes: The table fixes the within-nest electricity–capital elasticity, σ_{EK}, at the value in the first column and re-estimates the outer elasticity and capital-utilization parameter using the same identity-weighted minimum-distance criterion as the baseline model, subject to $\sigma_O \leq 1$. Predicted changes and RMSE are multiplied by 100 and are therefore reported in log points. A dagger indicates that the outer-elasticity estimate is at its upper boundary. The reduced-form targets equal the post-phase-out 2SLS coefficients evaluated at the 2010-total-salary-weighted mean supply shock. Generator owners retain their simulated grid-electricity response, while their non-electric inputs and output remain at baseline, as in the main simulation.

Under outer complementarity, the estimated outer elasticity reaches its upper boundary for the Leontief and 0.10 within-nest specifications, and these cases fit the reduced-form targets less closely. The estimates are interior for $\sigma_{EK} \in \{0.30, 0.50, 0.70\}$, with $\widehat{\sigma}_O$ declining from 0.88 to 0.45 and $\widehat{\kappa}$ remaining between 0.16 and 0.18. Among the displayed values, $\sigma_{EK} = 0.30$ provides the closest fit while preserving complementarity in both nests. Without the outer-complementarity restriction, the Leontief and 0.10 specifications instead require $\widehat{\sigma}_O = 1.23$ and 1.11, respectively. We therefore treat $\sigma_{EK} = 0.30$ as a representative nested-CES specification and interpret the remaining values as sensitivity checks.

E-5 Results for Elasticities

Table E4: Comparison of Elasticity Estimates

	Estimation method	
	Reduced-form (1)	Structural model (2)
Own-price elasticity (electricity)	-0.80	-0.74
Cross-price elasticity (labor)	-0.15	-0.15
Cross-price elasticity (materials)	-0.15	-0.15
Output elasticity	-0.20	-0.17
Preferred parameters	$\sigma = 0.57$, $\kappa = 0.17$	

Notes: The table reports elasticities of inputs and output with respect to electricity-price changes. Column (1) divides each main reduced-form IV coefficient by the estimated IV response of log electricity prices to the supply shock. This reduced-form ratio does not explicitly incorporate the model-implied shadow price of a binding rationing constraint: the outcome coefficient may reflect both price and rationing responses, whereas the denominator contains only the market-price response. Column (2) reports finite-change structural elasticities from the price-only simulation under $\widehat{\sigma} = 0.57$ and $\widehat{\kappa} = 0.17$. For each plant, we divide its predicted log outcome response by its predicted log market-price change and then take the 2010-total-salary-weighted mean; rationing and its shadow price are inactive by construction. Generator owners retain their simulated grid-electricity response, while their structural labor, materials, and output responses are set to zero.

F Counterfactual Analysis

F-0.1 Market integration and reserve sharing

Integrated-market pricing. We construct counterfactual electricity prices under market integration by specifying short-run regional supply curves, aggregating supply across regions, and solving for the integrated-market price. We parameterize the inverse supply curve in constant-elasticity form:

$$p_{rt} = \left(\frac{Q_{rt}}{c_{rt}} \right)^{\beta}, \tag{F1}$$

where p_{rt} is the electricity price, Q_{rt} is quantity supplied, $c_{rt} > 0$ is a region–year cost shifter, and $\beta \equiv d \ln p / d \ln Q$ (so $d \ln Q / d \ln p = 1/\beta$).[3]

Under full integration, all regions face a common price p_t^* in year t, implying regional supply

$$Q_{rt}^* = c_{rt} \, (p_t^*)^{1/\beta}. \tag{F2}$$

Holding fixed national demand $\bar{Q}_t$, the integrated equilibrium satisfies

$$\bar{Q}_t = \sum_r Q_{rt}^* = (p_t^*)^{1/\beta} \sum_r c_{rt}, \tag{F3}$$

which yields

$$p_t^* = \left(\frac{\bar{Q}_t}{\sum_r c_{rt}} \right)^{\beta}. \tag{F4}$$

Reserve sharing (effective capacity). Integration can reduce the need for region-specific reserve margins. We capture this mechanism by scaling aggregate effective capacity by $(1+\kappa)$:

$$p_t^* = \left(\frac{\bar{Q}_t}{(1+\kappa) \sum_r c_{rt}} \right)^{\beta}. \tag{F5}$$

We set $\kappa = 0.007$, corresponding to a 0.7% increase in effective capacity due to improved dispatch and reserve sharing.

[3] See Wolak (2000), Bushnell and Mansur (2005), Holland and Mansur (2006), and Cicala (2022).

Estimation. We set $\beta = 5$, corresponding to a supply elasticity $d \ln Q / d \ln p = 0.2$.[4] Rewrite (F1) in logs:

$$\ln p_{rt} = A_{rt} + \beta \ln Q_{rt}, \qquad A_{rt} \equiv -\frac{1}{\beta} \ln c_{rt}.$$

We recover $A_{rt} = \ln p_{rt} - \beta \ln Q_{rt}$ and model it as a function of generation shares and fixed effects:

$$A_{rt} = \gamma_1 Sh_{rt}^{\text{Nuclear}} + \gamma_2 Sh_{rt}^{\text{Hydro}} + \gamma_3 Sh_{rt}^{\text{Thermal}} + \gamma_4 Sh_{rt}^{\text{Others}} + \tau_t + \delta_r + \varepsilon_{rt}. \tag{F6}$$

We map the fitted value $\widehat{A}_{rt}$ back into the cost shifter via

$$\widehat{c}_{rt} = \exp\left(-\widehat{A}_{rt}/\beta\right), \tag{F7}$$

and substitute $\widehat{c}_{rt}$ into (F5) to obtain the integrated-market counterfactual price p_t^*.

Results. Table F1, Column (2), reports implied price changes under market integration. High-cost regions experience sizable counterfactual price reductions, while low-cost regions experience price increases, reflecting reallocation of generation toward lower-cost supply. The 2010-total-salary-weighted national average log-price change falls from 8.4 log points in the baseline to 5.7 log points under integration, corresponding to level increases of approximately 9.0% and 5.8%, respectively.

[4] Existing estimates of very short-run electricity supply elasticities are typically smaller (0.02–0.10) (Wolak, 2000; Bushnell and Mansur, 2005; Holland and Mansur, 2006; Cicala, 2022). Because our variation is at the region–year level, we adopt an intermediate value that allows for limited within-year adjustment.

Table F1: Regional Price Changes: Baseline and Market Integration

#	Region	$100 \times \Delta \ln p^E$	
		Baseline (1)	Market integration (2)
1	Hokkaido	1.1%	-12.6%
2	Tohoku	8.7%	1.7%
3	Tokyo	12.2%	3.1%
4	Hokuriku	4.3%	18.7%
5	Chubu	3.6%	5.0%
6	Kansai	12.9%	8.0%
7	Chugoku	3.6%	12.8%
8	Shikoku	5.0%	2.3%
9	Kyushu	6.4%	10.1%
	National	8.4%	5.7%

Notes: Entries report $100 \times \Delta \ln p^E$. Column (1) reports the shock-induced price increase (baseline), and Column (2) reports the counterfactual price change under market integration. National values are weighted averages over the current plant sample using 2010 total salary.

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